INCH-POUND

MIL-STD-3081
15 September 2026

DEPARTMENT OF DEFENSE
MANUFACTURING PROCESS STANDARD

WELDING OF ARMOR TITANIUM

AMSC N/A AREA THJM

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FOREWORD

1. This standard is approved for use by all Departments and Agencies of the Department of Defense.

2. This standard does not specify armor capability in terms of protection. The user is responsible for defining the threat that the armor system must withstand.

3. This is a new document without previous revisions.

4. Comments, suggestions, or questions on this document should be addressed to U.S. Army Ground Vehicle Systems Center, ATTN: FCDD-GVS-SAT MS #268, 6501 E. 11 Mile Road, Warren, MI 48397-5000, or emailed to usarmy.detroit.ccdc-gvsc.mbx.standardization@army.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

5. Questions or information with regards to this document related to ground vehicle systems should be addressed to gvscjoining@army.mil.

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1. SCOPE

1.1 Scope. This standard contains the minimum requirements for qualification of welding procedures, qualification of personnel, testing, fabrication, inspection, and repair of titanium alloy armor weldments. This standard is intended to be used to facilitate the fabrications of new weldments, repair of existing weldments, and govern the requirements associated with any weldment made in accordance with this document. The qualification tests required by this document are devised to demonstrate the adequacy of the welding procedures and to demonstrate the ability of welders or welding operators to produce sound welds. Do not use this standard for structural tubular T-Y-K- connections, pressure vessels, or media moving or process piping under high or low pressure.

2. APPLICABLE PUBLICATIONS

2.1 General. The publications listed in this section are specified in sections 3, 4, or 5 of this standard. This section does not include publications cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, users of this document are cautioned that they must meet all specified requirements of publications cited in sections 3, 4, or 5 of this standard, whether or not they are listed.

2.2 Government publications.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

TT-C-490	-	Chemical Conversion Coatings and Pretreatments for Metallic Substrates (Base for Organic Coatings)

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-46077	-	Armor Plate, Titanium Alloy, Weldable
MIL-DTL-32811	-	Titanium: Shapes, Forged or Formed; Bars, Flats, and Special Sections (for Armor)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916	-	DoD Preferred Methods for Acceptance of Product
MIL-STD-2035	-	Nondestructive Testing Acceptance Criteria

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DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-529	-	Use of Handheld Lasers to Remove Coatings and Corrosion from Aerospace Ground Equipment
MIL-HDBK-1222	-	Guide to the General Style and Format of U.S. Army Work Package Technical Manuals

(Copies of these documents are available from https://quicksearch.dla.mil/.)

2.2.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these publications are those cited in the solicitation or contract.

ARMY TECHNICAL MANUALS

TM 1-1500-335-23	-	Technical Manual Nondestructive Inspection Methods, Basic Theory
TM 1-1500-366-23	-	Technical Manual Work Package Nondestructive Inspection General Procedures and Process Controls

(Copies of these documents are available from https://armypubs.army.mil.)

ARMY TEST AND EVALUATION COMMAND

TOP-2-2-711	-	Proofing Projectile Impact Testing of Armor Weldments

(Copies of these documents are available from https://quicksearch.dla.mil/.)

NAVY TECHNICAL PUBLICATIONS

S9074-AQ-GIB-010/248	-	Requirements for Welding and Brazing Procedure and Performance Qualification

(Copies of these documents are available online via Model Based Product Support (MBPS) at https://mbps.navseaplm.navy.mil/Windchill/app/. To gain access to MBPS, obtain an account with National Help Desk Service Management (NHDSM) at https://nhdsm.navair.navy.mil (a valid CAC is required to access this website) and submit a SAAR-N Request. Refer questions, inquiries, or problems to (888) 292-5919.)

2.3 Non-Government publications. The following publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING

ANSI/ASNT CP-189	-	Qualification and Certification of Nondestructive Testing Personnel
ASNT SNT-TC-1A	-	Personnel Qualification and Certification in Nondestructive Testing

(Copies of this document are available online at https://www.asnt.org.)

ASTM INTERNATIONAL

ASTM E94/E94M	-	Standard Guide for Radiographic Examination Using Industrial Radiographic Film
ASTM E165/E165M	-	Standard Practice for Liquid Penetrant Testing for General Industry
ASTM E1032	-	Standard Practice for Radiographic Examination of Weldments Using Industrial X-Ray Film
ASTM E1254	-	Standard Guide for Storage of Radiographs and Unexposed Industrial Radiographic Films
ASTM E1417/E1417M	-	Standard Practice for Liquid Penetrant Testing
ASTM E1742/E1742M	-	Standard Practice for Radiographic Examination
ASTM E2699	-	Standard Practice for Digital Imaging and Communication in Nondestructive Evaluation (DICONDE) for Digital X-ray (DX) Test Methods

(Copies of these documents are available from https://www.aws.org/.)

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4	-	Standard Symbols for Welding, Brazing and Nondestructive Examination
AWS A3.0M/A3.0	-	Standard Welding Terms and Definitions Including Terms for Additive Manufacturing, Adhesive Bonding, Brazing, Soldering,

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		Thermal Cutting, Thermal Spraying, and Nondestructive Examination
AWS A5.02/A5.02M	-	Specification for Filler Metal Standard Sizes, Packaging, and Physical Attributes
AWS A5.12M/A5.12	-	Specification for Tungsten and Oxide Dispersed Tungsten Electrodes for Arc Welding and Cutting
AWS A5.16/A5.16M	-	Specification for Titanium and Titanium-Alloy Welding Electrodes and Rods
AWS A5.32M/A5.32	-	Welding Consumables – Gas Mixtures for Fusion Welding and Allied Processes
AWS B1.11M/B1.11	-	Guide for the Visual Examination of Welds
AWS B2.1/B2.1M	-	Specification for Welding Procedure and Performance Qualification
AWS B4.0	-	Standard Methods for Mechanical Testing of Welds
AWS B5.1	-	Specification for the Qualification of Welding Inspectors
AWS B5.2	-	Specification for Establishing a Company Program for Qualification and Certification of Welding Inspection Specialists
AWS C6.2/C6.2M	-	Specification for Rotary Friction Welding of Metals
AWS C7.4/C7.4M	-	Process Specification and Operator Qualification for Laser Beam Welding
AWS C7.6/C7.6M	-	Process Specification and Operator Qualification for Laser Hybrid Welding
AWS D1.9/D1.9M	-	Structural Welding Code – Titanium
AWS G2.4/G2.4M		Guide for the Fusion Welding of Titanium and Titanium Alloys.
AWS QC1	-	Specification for AWS Certification of Welding Inspectors
AWS WHB-1.10	-	Welding and Cutting Science and Technology VOLUME 1
AWS WHB-5.9	-	Welding Handbook Volume 5 - Materials and Applications Part 2

(Copies of these documents are available from https://www.aws.org/.)

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CANADIAN STANDARDS ASSOCIATION

CSA W178.2	-	Certification of Welding Inspectors

(Copies of this document are available online at https://www.csagroup.org/.)

COMPRESSED GAS ASSOCIATION

CGA E-1	-	Standard for Rubber Welding Hose and Hose Connections for Gas Welding, Cutting, and Allied Processes

(Copies of this document are available online at https://www.cganet.com/.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO/IEC 17025	-	General Requirements for the Competence of Testing and Calibration Laboratories
ISO 3821	-	Gas welding equipment — Rubber hoses for welding, cutting and allied processes

(Copies of this document are available online at www.iso.org.)

NATIONAL AEROSPACE STANDARD

AIA/NAS410	-	NAS Certification & Qualification of Nondestructive Test Personnel

(Copies of this document are available online at https://www.aia-aerospace.org/.)

SAE INTERNATIONAL STANDARD

SAE AMS2814	-	Packaging and Marking of Packages of Welding Wire Premium Quality
SAE AMS4954	-	Titanium Alloy, Welding Wire 6Al - 4V - (Composition similar to UNS R56402)
SAE AMS4956	-	Titanium Alloy Welding Wire 6Al - 4V, Extra Low Interstitial Environment Controlled Packaging - (Composition similar to UNS R56408)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. DEFINITIONS

3.1 Active supervision. Direct on-site readily available supervision supplied by the certified visual inspector (CVI) (see 5.8.2.1) in the form of necessary instructions, into which ensure the authorized visual inspectors (AVI) (see 5.8.2.2) can execute specific inspections (see AWS QC1).

3.2 Contractor. The organization having a direct contract with the United States Government. In this standard, the term contractor refers to the prime contractor.

3.3 Cyclic load. A load that is being repeatedly applied and removed during the lifespan of the weldment.

3.4 Deposition thickness. The depth of weld metal deposited in a weld joint for each process, transfer mode, or filler material classification used. This term is associated with single and multipass welds, complete and partial penetration joints, and multi-process welds.

3.5 Design engineering activity. The authorized personnel with overall design authority of the system, sub-system, or component.

3.6 Digital radiography. All radiography methods whereby images are in a digital format.

3.7 Entity. All sites of an organization under the same quality assurance management plan and using the same quality manual, performing work to which this document is applicable.

3.8 Entity responsible representative. Any representative of an entity that is specifically authorized to approve equipment, material, or procedures within the scope of this document.

3.9 Essential variables. Essential variables are those elements, either material or process, that are important in establishing a welding procedure. These variables are defined as part of a welding procedure. Changes in these variables, after a procedure has been qualified, require a change in the written procedure and may require requalification, depending on the element being changed and the magnitude of the change based on the requirements of this document.

3.10 Field repair. Reprocessing of nonconforming material in accordance with approved written repair procedures and operations to address the nonconformity in theater. The purpose of a field repair is to restore functionality to an item being repaired on an as-needed basis. Field repairs are distinguished from other repairs, in that the item, after repair, is temporarily excused from completely conforming to all of the applicable drawings, specifications or contract requirements. The system will be brought back into full conformity once the repaired item reaches a facility that is capable of performing a complete repair. This type of repair is not to be

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used in manufacturing of new vehicles or repairs at a facility capable of performing a complete repair.

3.11 Forced cooling. The cooling of welds using any other method greater than 5 mph air movement at ambient air temperature.

3.12 Hot pass. Process conducted at elevated temperature between welding passes.

3.13 Hybrid welding. A combination of two or more welding processes concurrently applied to generate a weld bead.

3.14 Inadvertent inspection. Unplanned or unintentional observed indications or defects discovered during NDT processing of nearby areas.

3.15 Manufacturer. The entity performing the operations covered by this standard. The contractor may or may not be the manufacturer. The manufacturer may also be known as the sub-contractor or sub-contracting entity.

3.16 Material review board (MRB). A board consisting of contractor and government personnel necessary to review, evaluate, and determine or recommend disposition of nonconforming material referred to it.

3.17 Nominal thickness. The thickness that is specified on the plans or drawings without application of any allowed tolerance. The abbreviation for nominal thickness is T. Tolerances for nominal thicknesses will be dictated by the associated base material specification.

3.18 Procuring contracting officer (PCO). The individual authorized on behalf of the government, and who is responsible for overall procurement under the contract.

3.19 Proofing projectile impact testing (PPIT). This test was previously referred to as the ballistic shock test (BST). This is a high velocity impact test performed with a blunt nose proofing projectile to evaluate the performance of the weld under this type of impact. This test is performed in accordance with TOP-2-2-711, ballistic testing of armor weldments. This test is not correlated to system level performance specifications or other system requirements.

3.20 Reference standard. A material or object for which all relevant chemical and physical characteristics are known and measurable, used as a comparison for, or standardization of, equipment or instruments used for nondestructive testing.

3.21 Repair. Reprocessing of defective welds, discontinuities, adjacent welding regions, base material, and improperly located components or weldment features; as to subsequently bring into conformance with this document, drawings, specifications, and contract requirements.

3.22 Repair procedure. A unique document that works in conjunction with a qualified weld procedure specification (WPS) to provide a detailed plan for each area that requires repair.

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3.23 Standard welding terms and definitions. All standard welding terms and definitions are located in the latest edition of AWS A3.0M/A3.0.

3.24 Technical authority (TA). Entity responsible to establish, approve, and assess conformance to technical, safety, and certification requirements and policy for both products and processes within this document. The term TA term can refer to the overall TA for a program or the TA for a specific engineering or technical discipline or design consideration. For example, the government welding engineer is the TA for welding procedure qualification review and the design engineer is the TA for joint design classification.

3.25 Validation plate. Test plate for demonstration of unchanged mechanical properties following the change of cleaning process, transfer of procedure, or other alteration of a previously qualified procedure.

3.26 Welder qualification continuity record. A document or database that contains the record of the welder or welding operators training and usage of welding processes and procedures they are qualified to perform. The record shows continuity of usage of the qualified process for each welder or welding operator and notes the date the process must be used before expiration of qualification.

3.27 Welding engineer. The Government or entity's engineer assuming technical authority for welding engineering responsibilities, including but not limited to, weld design, quality, procedure and personnel qualification and development.

3.28 Welding equipment. All non-consumable equipment used for, or in support of, welding, and that provides instantaneous information to the user about the welding process. Information can be in the form of data output such as the amperage reading on the welding machine, or temperature measuring devices or gauge readings such as flow meters for shielding gases.

3.29 Welding symbols. Welding symbols are to be those shown in the latest edition of AWS A2.4.

4. GENERAL REQUIREMENTS

4.1 Qualification requirements. All entities shall qualify procedures and personnel in accordance with this standard and submit the applicable documents (see 5.3, 5.5, 5.6, and 5.8) to the procuring contract officer upon request, and prior to welding any production weldments (see 6.2). This document shall serve as the fabrication document, when a fabrication document is referenced to in any other document, unless otherwise required by the procuring contract officer.

4.1.1 Submission format. Procedure qualification records (PQRs), weld procedure specifications (WPSs), and welder qualification records (WQRs) shall be in an organized and efficient packet. Guidance and overview in the form of a supplementary table of contents, outlines, tables, charts, or matrices are encouraged. The quantity of each said document should be identified in the submission along with number of pages. Illegible or blurry submissions are unacceptable and shall be re-produced if clarity cannot be met.

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4.1.1.1 PQR submission format. Each PQR shall be uniquely identified with a traceable identifier and submitted to the procuring contract officer with all applicable supporting documentation upon request. Supporting documentation for PQRs includes but is not limited to, nondestructive and destructive test data and results, filler material certifications, lot conformance testing results, shielding certifications, shot records, and any revision of the PQR (see 5.3, 5.4, 5.7 and 5.9). PQR and supporting documentation shall be put into a packet and delivered for all applicable requests.

4.1.1.2 WPS submission format. Each WPS shall be uniquely identified with a traceable identifier and submitted to the procuring contract officer with all applicable supporting documentation upon request. Supporting documentation for WPSs includes but is not limited to supporting PQRs, shot records, previous revisions of WPSs or PQRs, and testing data/results (see 5.4, 5.5, 5.7, and 5.9). Each WPS and supporting documentation shall be put into a packet and delivered for all applicable requests.

4.1.1.3 WQR submission format. Each WQR shall be uniquely identified with a traceable identifier and submitted to the procuring contract officer with all applicable supporting documentation upon request. Supporting documentation for WQRs includes but is not limited to nondestructive testing results, WPS used for WQR Test, and destructive testing results (see 5.6, 5.7, and 5.9). WQR and supporting documentation shall be put into a packet and delivered for all applicable requests.

4.2 Qualification applicability. Qualifications performed and approved in accordance with this document are intended to be based on material type and weld classification and are applicable to any weldment that aligns with the scope of the qualification.

4.3 Responsibility. When this document is specified by the contract or drawing, each entity qualifying procedures under the requirements of this document shall prepare written welding procedures and perform the required tests to qualify these procedures. Prior to any fabrication, the entity shall produce all PQRs, WPSs, and WQRs for all welds in accordance with this document and shall be delivered to the procuring contract officer upon request. Examples of a blank PQR, WPS, and WQR are in FIGURE B-1, FIGURE B-2, and FIGURE B-3 respectively.

4.4 Previous qualification to AWS D1.9/D1.9M Annex A. Procedures and personnel previously qualified, for which records are available, and qualifications have been maintained (see 5.3.7, 5.5.6, and 5.6.9), are acceptable for use in this document. All previously approved welding procedures and personnel qualifications, and their supporting documentation shall be submitted to the procuring contract officer and the technical authority for approval for each scope of work upon request (see 6.2). TABLE I shall be used as a reference to determine the new weld classification (see 4.6.1). The technical authority may require requalification to this document on a case-by-case basis where any of the following apply:

a. There are specific concerns about the adequacy or accuracy of the welding procedures, the ability of the personnel to produce acceptable works, or which are not able to meet the requirements of the applicable fabrication document.

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b. The existing qualification does not meet the requirements of its original qualification document.

c. Any exceptions made to a previous procedure that does not meet the requirements of the original qualification document.

4.5 Safety precautions. This standard may involve hazardous materials, operations, and equipment. This standard does not address any of the safety issues associated with their use. It is the responsibility of the user to establish appropriate health and safety practices.

4.6 Joint design. The weld joint design shall be in accordance with engineering best practices and shall be the responsibility of the entity's design engineering activity with the approval of the procuring contract officer upon request. All joint designs and joint design calculations, including finite element analysis (FEA) results, shall be delivered to the PCO for review upon request. Joint designs incorporating pipe or tube shall be non-ballistic and shall be in accordance with AWS D1.9/D1.9M.

4.6.1 Weld joint classification requirements. Each weld made using this standard shall be classified based on the service conditions. The classification governs the extent to which quality assurance provisions are applied to the weld joint. The design engineering activity shall determine the classification of welds in accordance with this document and TABLE II with final weld classification approval by the procuring activity. Factors to consider when classifying welds include: high-strain-rate events, static loading versus dynamic loading, cyclic loading, corrosion, temperature, material characteristics, probability of failure, consequences of failure. Where a weld can be assigned to more than one class, the more stringent class shall be selected. Where welds intersect or overlap another weld of a higher classification, the intersection point and 2 inches (50.8 mm) in all directions from the intersection point shall meet the quality assurance requirements of the higher classification (see FIGURE 1). Note that weld joint classification is not related to base material classes for armor grade titaniums, system level performance specification requirements or NDT acceptance criteria. Weld classes shall be chosen on the basis of the following definitions:

a. Class I welds. Applies to welds in critical load bearing elements that are not fail-safe and are constructed of armor grade titanium. Class I welds are typically used in primary load bearing connections and for survivability. Failure of a Class I weld in service would be catastrophic and would result in the loss of life, system(s), control, major components, unintentional release of critical stores, or a mission-critical failure. A Class I Weld shall be categorized in accordance with TABLE II. Class I welds shall be qualified in accordance with this document. A Class I weld shall be categorized as "High" on the weld classification decision matrix. Possible examples of Class I welds include but are not limited to: turret structure, armor solution, monocoque armored hull, and exterior components and assemblies.

b. Class II welds. Applies to welds in cyclic or static load bearing elements that are either fail-safe or not fail-safe and are constructed of any combination of titanium requiring greater levels of inspection. Class II welds are typically used in secondary load bearing (i.e., shared load) connections. Failure of a Class II weld in service would reduce the overall

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efficiency of the system, and may result in the loss of life, system(s), control, major components, or a mission-critical failure. A Class II weld shall be categorized in accordance with TABLE II. Class II welds shall be qualified as cyclic and static load bearing welds (Class A) in accordance with AWS D1.9/D1.9M. Possible examples of Class II welds may include but are not limited to: structural welds, attachments, interior components, door or hatch components, leak-proof containers, ammunition racks, and moderately critical components.

c. Class III welds. Applies to welds that are in static load bearing and noncritical elements and are constructed of any combination of titanium. Failure of Class III welds would have minor to no effect on the efficiency of a system and endangerment to personnel may occur. Class III welds are typically used in connections where any expected load transfer at the weld would be a secondary or tertiary load bearing (i.e., shared load) connection or negligible. A Class III Weld shall be categorized in accordance with TABLE II. Class III shall not be specified for welds used for making connections onto critical or primary load path elements (e.g., lift points, etc.) or elements directly related to personnel supporting activities, regardless of the loading condition/direction. Class III welds shall be qualified as a static load bearing welds (Class B) in accordance with AWS D1.9/D1.9M. Examples of Class III welds include but are not limited to: ladders, handrails, stowage boxes, attachments, brackets, fenders, heaters, air ducts, boiler casings, air vents, and most types of sheet metal assemblies, and non-critical components.

4.6.1.1 Armor welds without class designations. When a weldment is constructed of an armor grade titanium welded to an armor grade titanium, or to a titanium approved for use as an armor solution, and does not have a weld classification, the technical authority shall assign the weld classification. The procuring contract office shall, in its best effort, contact the original design activity to define the susceptibility and severity of weld failure for this particular weldment. If the original design activity is unavailable, the technical authority shall work with the design engineering activity to define weld classification for this particular weldment. When a weldment consists of armor grade titanium welded to non-armor grade titanium, or non-armor grade titanium welded together, that does not have a weld classification, the design engineering activity shall determine the classification of welds in accordance with this document.

4.6.1.2 Welding symbol requirements for classification. To minimize fabrication costs by avoiding over-inspection and unnecessary repair, all welds on a drawing shall be classified individually. This can be accomplished by including a note on the engineering drawing with the general format shown in FIGURE 2 which specifies only the DoD nomenclature. The weld class shall then be indicated by one the following:

a. Detailing the specific weld class within the tail of welding symbol for each individual weld joint, or

b. By using specific flag notes with the welding symbol for the individual weld joints (see FIGURE 3) or,

c. By using detailed drawing notes, provided that all of the welds on each detail drawing are the same weld class.

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4.6.2 Fillet weld size. Unless specified in drawings or contract requirements, all fillet welds shall conform to the requirements listed in AWS D1.9/D1.9M.

4.6.3 Intermittent fillet welds. Intermittent fillet welds shall not be allowed for Class I. For all other classes of welds, length and pitch specified shall be a result of design calculations and shall conform to the requirements listed in the Minimum Length of Intermittent Fillet Weld section of AWS D1.9/D1.9M.

4.7 Welding processes.

4.7.1 Acceptable welding processes. The following welding processes are acceptable for use in accordance with this document without restriction.

a. Gas Metal Arc Welding (GMAW)

b. Gas Tungsten Arc Welding (GTAW)

4.7.2 Restricted welding process. The following welding processes listed below shall require with approval from the technical authority with supporting data.

a. Plasma Arc Welding (PAW)

b. Electron Beam Welding (EBW)

4.7.3 Additional welding processes. Additional welding processes not covered by this document shall require qualification in accordance with the document listed within this section for each process below. Each welding process shall be identified on the engineering drawing. Use for Class 1 welds (see 4.6.1) shall require ballistic testing (see Appendix A) and destructive testing (see 5.7) in accordance with this standard.

a. Stud Welding (SW) – S9074-AQ-GIB-010/248.

b. Rotary Friction Welding (RFW) – AWS C6.2/C6.2M

c. Laser Beam Welding (LBW) – AWS C7.4/C7.4M

d. Laser Hybrid Welding (LHW) – AWS C7.6/C7.6M

e. Other Welding Processes – Unlisted welding processes shall qualify via applicable industry standards. Other welding processes may be used with approval of the procuring contract officer and the technical authority (see 6.2). Additional welding and workmanship requirements may be required, depending on the process to be approved. For additional guidance see AWS G2.4/G2.4M.

4.8 Weld positions. For descriptions of all welding positions and position limitations, see the most recent version of AWS A3.0M/A3.0.

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4.9 Quality control. It shall be the full responsibility of the entity to institute and maintain a quality program and the inspection standards necessary to ensure that the part, assembly, sub-assembly, and end product is produced and meets the requirements of the contract, drawings, applicable regulations, system relevant technical manuals and bulletins, and this document. The quality program shall also ensure that measuring, test and manufacturing equipment are properly maintained, and that appropriate actions are taken for nonconforming materials.

4.9.1 Quality assurance system. Each entity shall maintain a quality assurance system to assure the procuring contract officer that all requirements of this document have been and are continuously being met. Written procedures shall be prepared to assign responsibility and provide accountability for performing work and inspections. This system shall be defined in a document such as the entity's quality document and shall be made available for review to the PCO upon request.

4.9.1.1 WPS transfer requirement. A section in the quality manual specifically addressing the transfer of qualified procedures from one site of an entity to another site within the same entity. For requirements to transfer from entity to sub-entity (see 5.5.3.2).

4.9.2 Welding equipment calibration and verification. Equipment calibration shall be established and verified to a calibrated meter at a minimum of an annual basis. The entity shall develop and maintain an equipment verification program for all welding equipment. All equipment verifications shall be performed with a calibrated meter that is NIST certified and documented within the entity's quality assurance system.

4.10 Welding prohibitions. Unless specifically approved by the technical authority, the following are prohibited:

a. Welding in the vertical down position.

b. Anti-spatter dip or spray

c. Welding over coated surfaces, such as primers, and paint (see 5.11.1).

d. Welding on material when water, oil, or similar materials are in contact with the surface opposite the side to be welded and less than 1/4 inch (6 mm) of base material is between the weld area and the surface in contact with the liquid, or when base metal is below the dewpoint.

e. Welding without an inert gas or vacuum shield

f. Welding without cleaning

4.11 Company name change. Properly documented WPSs and PQRs qualified under the provisions of this standard by a company that later has a name change due to voluntary action or consolidation with a parent company may utilize the new name on its WPS documents while maintaining the supporting PQR qualification records with the old company name.

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4.12 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

5. DETAILED REQUIREMENTS

5.1 Base material.

5.1.1 Base material certification. The material certification from the manufacturer shall be verified by the contractor against the requirements of the applicable specification prior to its usage. All certification documentation of base material, as outlined in this section, shall be accessible by the entity according to record retention requirements (see 5.1.8).

5.1.1.1 Base material control requirements. The identification of the material shall be maintained to the point of initial fabrication in accordance with a written procedure to ensure traceability. The identification of the material shall be verified at the point of initial fabrication as meeting the requirements of the drawing.

5.1.2 Armor base material. All accepted titanium alloy armor shall be limited to materials qualified under MIL-DTL-46077 or MIL-DTL-32811. For material qualified under MIL-DTL-32811, the equivalent plate class in MIL-DTL-46077 shall be used to qualify welding procedures. Alignment of armor class to M-number is provided in table III for information only.

5.1.3 Base material grouping. Base materials groupings for armor shall be prohibited. Base material groupings for non-armor materials shall be in accordance with AWS D1.9/D1.9M base metal section, AWS B2.1/B2.1M base metal grouping section.

5.1.4 Dissimilar base materials. Joints consisting of two different base material groups shall be tested as a plate with each base material group (see 5.1.3). Base material groupings shall be prohibited for armor. Additionally, different chemistries that have been qualified under MIL-DTL-46077 Class 4 or MIL-DTL-32811 Class IV shall be treated as separate grades. A procedure that has been qualified to each base material shall not be automatically be qualified for a joint of dissimilar base materials or base material groups.

5.1.5 Backing material.

5.1.5.1 Permanent backing material. Permanent backing for ballistic welds shall not be permitted unless detailed on the drawing. Permanent backing shall be held to the grouping and dissimilar base material requirements of 5.1.3 and 5.1.4.

5.1.5.2 Removable backing material. If using removable backing material, a change in backing material type, unless to a material that is the same as one of the base materials or the weld filler material, shall be cause for requalification (see 5.4.2.4).

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5.1.6 Spacer material. Spacers shall be fabricated from the same grade of materials as the base material.

5.1.7 Run off tab material. Run off tab material shall be removed at the completion of welding before NDT or visual inspection. Run off tabs are acceptable for use with all classes of welds.

5.1.8 Material record retention. Material certifications for all armor material as provided by the mill shall be accessible for the period of performance of the contract at a minimum and be delivered to the procuring contract officer upon request.

5.2 Welding materials.

5.2.1 Identification. All welding filler and shielding materials shall be identified up to the point of usage.

5.2.1.1 Electrodes. Each spool or coil of electrodes shall carry an identifying label. Each piece of filler metal shall have distinguishable color code, type designation, or classification number marking.

5.2.1.2 Shielding gas. Each shielding gas line shall be identified near the welding machine as to type and composition.

5.2.2 Filler materials.

5.2.2.1 Requirements. Grades of filler materials that may be used for the corresponding welding processes and base materials are listed in TABLE IV. Filler material sizes, packaging, and physical attributes shall be in accordance with AWS A5.02/A5.02M or SAE AMS2814. The use of these filler materials does not guarantee successful process qualification testing. Any filler material not listed may be used if all procedure qualification criteria are met in accordance with this standard.

5.2.2.2 Filler material control requirements. The entity shall have a consumables management procedure including but not limited to; identification, storage, handling, preservation, inventory control, and distribution.

5.2.2.2.1 Filler material handling. Welding electrodes shall be handled carefully to prevent damage to their coatings, if coatings exist. Damaged containers shall have their contents examined for excessive moisture content or other damage to the electrodes. Filler spools or drums shall be examined for distorted spools, entwined winding on spooled wire, or surface contaminates. Filler material damaged to the extent that it does not meet specification requirements shall not be used for production welding.

5.2.2.2.2 Filler material inventory control. All welding materials shall be identified up to the point of usage. If markings are destroyed by baking or other means, electrodes shall not be used until identification is restored. Each spool or coil of bare filler metal shall carry an identifying label. Cleaned filler material may be stored in a nonpermeable bag or container that is

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certified to maintain cleanliness of the weld filler material until it is opened and ready for use. Weld filler material shall be stored in a clean, dry environment, preferably in a storage container or bag. Introducing inert gas into the bag or container may be used to control the oxidation rate to keep the filler metal from re-oxidizing.

5.2.2.3 Filler material record retention. Filler material certifications shall be kept for the period of performance of the contract at a minimum and delivered to the procuring contract officer upon request.

5.2.3 Shielding.

5.2.3.1 Shielding gas. All shielding gas or gas mixtures shall be in accordance with AWS A5.32M/A5.32 and shall be certified by the gas manufacturer. A gas or gas mixture used for shielding shall be of a welding grade and shall have a dew point in accordance with the "Minimum requirements on purities and moisture contents of gases and gas mixtures" table of AWS A5.32M/A5.32. When shielding gas mixers are required, calibrated mass flow meters shall be used for proportioning the percentages of gases to conform to the requirements of the WPS (see 4.9). The procuring contract officer may conduct reviews of the calibration and certification documentation to assess conformance to this standard.

5.2.3.2 Purging gas. All purging gas or gas mixtures shall be in accordance with shielding gas requirements (see 5.2.3.1).

5.2.3.3 Shielding record retention. All shielding certifications from the manufacturer shall be kept by the entity for the period of performance of the contract at a minimum and be delivered to the procuring contract officer upon request.

5.3 Procedure Qualification Records (PQRs).

5.3.1 PQR requirements**.** This section describes the general requirements for developing and qualifying a PQR. A welding PQR shall be prepared in accordance with the requirements of this document prior to any welding, except for welding required to support the development of the PQR. The PQR shall list applicable procedure qualification variables specified in TABLE V for the class of weld being qualified. Data entered on the PQR shall be the actual item description or numerical value recorded during the welding of the procedure qualification test assembly. All observed values shall be recorded as single data points for each variable used during welding. For inherently fluctuating variables, use the mean value observed during welding. The PQR shall be developed and serve as the basis for the welding procedure specification (WPS). The PQR shall include, as a minimum welding parameters used for the welding of the qualification test assembly, destructive testing (see 5.7) results, and NDT examination (see 5.8). All supporting testing records shall be included with the PQR. Qualified WPSs, along with supporting PQR(s) (with supporting testing records) shall be submitted to the procuring contract officer upon request (see 6.2). PQRs shall be signed by the entity's responsible representative stating that the PQR and all testing results conform to the requirements of this standard.

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5.3.1.1 PQR requirements for ballistic welds. All ballistic PQRs shall require a qualification test plate to be fabricated in accordance with FIGURE 4 for destructive testing in accordance with TABLE VI, TABLE VII, and 5.7, and an I-plate in accordance with TABLE VI, TABLE A-I, and FIGURE A-1 for proofing projectile impact testing. All applicable testing and essential variables of TABLE V shall be met for the PQR to be qualified. A weld procedure qualification performed in this manner qualifies the procedure for all weld classes.

5.3.1.2 PQR Requirements for non-ballistic welds. All Class II PQRs shall be qualified in accordance with AWS D1.9/D1.9M as cyclically loaded and critical statically loaded structures (Class A). All Class III PQRs shall be qualified in accordance with AWS D1.9/D1.9M as statically loaded structures (Class B). All applicable testing and essential variables shall be met for the PQR to be qualified. A Class II PQR, comprised of armor titanium to armor titanium configuration, can be developed into a Class I PQR in accordance with 5.3.3, after developing and successful testing of a supplemental I-plate in accordance with TABLE VI, TABLE A-I, and FIGURE A-1 for PPIT.

5.3.1.3 PJP and fillet groove weld PQRs. Any PJP qualification, including fillet welds, involving non-armor to non-armor materials shall be qualified in accordance with AWS D1.9/D1.9M or the following requirements. Any PJP qualification involving armor, including fillet welds, shall require a CJP groove weld qualification test prior to performing the PJP or fillet weld qualification.

5.3.1.3.1 PJP groove welds. A supporting CJP PQR shall serve as the base mechanical testing qualification for any associated PJP groove welds. In addition, three macrographs shall be required for each groove weld type to validate sufficient penetration depth, weld size, and proper fusion has been achieved in the PJP configuration. If a Class I PJP is desired, the PQR shall be qualified in accordance with 5.3.1.1 in addition to the requirements of this paragraph.

5.3.1.3.2 Fillet welds. Fillet weld qualification shall require both single pass and multiple pass fillet welds as required for production to determine maximum single pass weld size and minimum multipass weld size in accordance with TABLE VIII. If multipass is not required for production, single pass may be qualified independently. Fillet welds for non-armor materials shall be qualified in accordance with AWS D1.9/D1.9M. Macrograph tests shall be required in accordance with TABLE VI and FIGURE 5 to validate root penetration, root fusion, leg fusion, weld leg size, and actual throat achieved in the fillet configuration.

5.3.1.4 Multiple welding processes used in PQR. A PQR test weld assembly that requires multiple passes by way of using different welding processes shall have all essential variables of each process listed in the PQR data sheets. If multiple processes are intended for use in production of a single joint application, then all of the processes shall be qualified in a single PQR test assembly. The deposition thickness (t) of each process to be qualified in a single joint shall be based on the thickness deposited with each process, and qualified thickness for all processes combined in one joint shall be based on the PQR test assembly thickness (T) (see TABLE IX). A single process may be utilized from this type of qualified procedure, provided that all limitations of that process are observed. However, the root layer(s) up to a thickness of 3/16 inch (4.7mm) shall be deposited with the same process used in the qualification of the root layer(s).

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Each process shall be included in all destructive test specimens. Where specimens are too small to include representative weld metal from all processes, all required test specimens shall be taken from weld metal from each process. An exception shall be made for a process subject to the following limitations:

a. The weld process involves no more than two layers in qualification and production welds (e.g., the root plus one layer, etc.), and

b. The test specimens involved cannot be removed so as to include this limited weld region.

Example – when welding a 1 inch (25.4 mm) thick PQR test plate (T), GTAW was used for the root with GMAW balance used for the remaining fill and cover passes. 1/4-inch (t) GTAW and 3/4-inch (t) GMAW were deposited. Therefore 1/2 inch (2t) for GTAW and 1 ½-inch (2t) for GMAW are qualified (see TABLE IX). However, GMAW is not qualified to weld the root of the joint because it was not qualified in the PQR for root pass welding.

5.3.2 PQR revision. Revision of PQRs shall be prohibited, except for minor spelling and grammatical errors.

5.3.3 Supplemental PQRs. A supplemental PQR is a PQR that is developed to support or expand the parameters of a WPS. Supplemental PQRs shall be allowed to be created and incorporated as additional supporting PQRs to an existing qualified WPS for all classes of welds. To create a Supplemental PQR, a test plate shall be created in accordance with the requirements of 5.3.1 and its applicable subparagraphs. The welded supplemental test plate shall then be subjected to all of the required nondestructive testing (see 5.9) and the lacking destructive testing (see 5.7) as required to obtain the qualification. Dependent on the weld classification a PPIT (see Appendix A) may also be required to obtain qualification. All required test results shall be listed on the supplemental PQR. A supplemental PQR may include the following items absent from the original PQR:

a. Qualification data that is required by this document such as destructive testing results as required by 5.3, 5.4, TABLE V, and the required supporting documentation.

b. For upgrading to Class I welds, PPIT results performed in accordance with Appendix A and the required supporting documentation.

c. All essential variables listed within this standard that are not present on the original PQR (see 5.3, 5.4, and TABLE V).

A supplemental PQR can allow for a previously qualified WPSs of Class II or Class III welds to now attain Class I qualification. Previously qualified procedures are addressed in 5.3.4. For procedures where proofing projectile impact testing data is available, but exceptions were granted from the original qualification document, a supplemental PQR shall be developed to attain qualification in accordance with this document.

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5.3.4 Previous qualified welding procedures. Previously qualified welding procedures shall meet all applicable essential variables of AWS D1.9/D1.9M. If the previously qualified welding procedure essential variables do not meet the requirements of the original qualification document, or any of the other requirements of 4.4, then a new PQR shall be created, in accordance with this document. If the essential variables are met but PPIT results are not available, a new PQR shall be created in accordance with this document.

5.3.5 Bead width limitation for weaving. To qualify the use of a weaving technique, a PQR using a weaving technique shall be generated. The maximum weave width to be used in the WPS shall be limited to the maximum weave width qualified in the PQR test plate.

5.3.6 Base material repair qualification.

5.3.6.1 Armor base material repair applications For repair qualifications of armor plate base material, a simulated repair plate shall be prepared using a qualified single sided WPS for each position of welding in which the procedure application is intended. The WPS shall be single sided, and conform to the weld thickness ranges qualified in TABLE IX. The qualified WPS utilized for the repair may be open root, have a backing solution, and or backing gas and be used on this closed root weld repair. All repairs shall maintain a minimum of 1/8" of remaining root thickness minimum. Backing gas, backing seal, or with backing, shall conform to the WPS solution until the root thickness exceeds ½", at which point no backing solution is needed. The weld thickness is the depth of the repair. The WPS shall have completed destructive testing in accordance with 5.7. The dimensions of the plate shall be 24 in. x 24 in. (610 mm x 610 mm), at a minimum, but shall be of sufficient size to allow for a minimum of 5 simulated repairs. Each desired weld thickness grouping, (see TABLE IX) shall be tested. Multiple weld thicknesses may be tested on 1 plate. In addition, the condition of the final repair weld surface should be the same as that intended for production application, i.e., as contoured by welding or mechanical finishing. The simulated repair plate shall pass radiographic inspection requirements (see 5.9.2.3). The simulated repair plate shall then be subjected to ballistic testing in accordance with the applicable MIL-DTL ballistic tests section to determine compliance with the requirements of the minimum required ballistic limits of the plate thickness. The simulated repair shall meet or exceed the required ballistic limits for all relevant threats of the MIL-DTL to be qualified as an armor base material repair procedure. Procuring contract officer approval of the listed qualification documents and a proposed detailed repair procedure (see 5.12.2) shall constitute approval for repair.

5.3.6.2 Non-armor base material repair applications. A CJP groove welding procedure qualification involving the same base material group or groups to be repaired shall constitute approval for repair and build-up. The deposition thickness qualified in the WPS shall limit the maximum qualified thickness to be deposited for the repair. The filler material to be used for the repair shall match the, as welded, mechanical properties of the base material to be repaired. Post weld treatment may be required to achieve required properties.

5.3.7 PQR record retention. All PQRs shall be identified in a manner to allow permanent traceability. PQRs shall be retained as long as the PQR is applicable. Documentation of the test results detailed in the procedure qualification shall be kept by the entity as long as the

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PQR is applicable. All PQRs and testing data shall be delivered to the procuring contract officer upon request.

5.4 Essential variables.

5.4.1 Essential variables. This section describes the essential variables for PQRs and WPSs. The PQR essential variables are detailed in this section and TABLE V. Each applicable variable shall be determined for each class of weld, recorded as a single variable on the PQR data sheet for each weld pass, and meet the requirements of this section and TABLE V. See 5.3 for further details on PQR development. The WPS essential variables and their ranges are detailed in this section and TABLE V. These essential variable ranges shall be applied to the WPS (see 5.5).

5.4.2 Base material essential variables.

5.4.2.1 Base material specification . A change in class or grade outside of permitted material groupings for welding procedures in accordance with 5.1.3 shall be cause for requalification. A change in material form shall be permitted so long as all chemical requirements remain the same.

5.4.2.2 Base material thickness (T). A change in base material thickness outside the limits for which the procedure was qualified shall be a cause for requalification (see TABLE IX).

5.4.2.3 Deposition thickness (t). The thicknesses of each process, transfer mode, or filler material classification used shall be recorded in the PQR. A change exceeding the qualified maximum weld deposit thickness is a cause for requalification.

5.4.2.4 Backing or spacer material. A change of material grade of the backing or spacer strip shall be cause for requalification (see 5.1.5). A change in adhesion method shall be cause for requalification. A greater than 5% reduction in width of the backing or spacer strip shall require the preparation of a validation test plate (see 5.3.3) to demonstrate the maintenance of alpha-phase volume fraction.

5.4.3 Filler material essential variables.

5.4.3.1 Electrode and rod classification. A change in electrode or rod classification, except as permitted by TABLE IV shall be cause for requalification.

5.4.3.2 Number of electrodes. Any change in the number of electrodes used shall be cause for requalification. This includes autogenous welds, made without filler material.

5.4.3.3 Cold wire and hot wire. A change from cold wire to hot wire or vice versa shall be cause for requalification.

5.4.3.4 Electrode diameter. A change in electrode diameter outside of the following limitations:

a. For GMAW, any increase or decrease of 1/64 inches (0.4 mm)

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b. For GTAW, any increase or decrease of 1/16 inches (1.6 mm)

5.4.4 Shielding.

5.4.4.1 Flow rate. A change in gas flow rate with an increase of 50% or a decrease of 20% shall be cause for requalification of a procedure. If shielding fails in process, immediate visual inspection shall be conducted prior to continuance of welding or corrective action (see 5.9.2.1).

5.4.4.2 Composition. A change from one single gas to another single gas or to a gas mixture shall be cause for requalification. Any change in the specified nominal percentage of a gas mixture shall be in accordance with AWS A5.32M/A5.32. A change in gas mixture outside of AWS A5.32M/A5.32 shall be cause for requalification. Analytical equipment for monitoring gases shall be capable of measuring oxygen levels to better than 10ppm.

5.4.4.3 Gas hose selection. A change from one flexible hose material to another shall require verification of moisture level at the point of welding prior to production. Hoses made in accordance with ISO 3821, CGA E-1, or another standard may be exempt from this requirement if documentation is delivered to the procuring contract officer for approval upon request (see 6.2).

5.4.4.4 Trailing shield. Change in shielding method from trailing shield to inert or vacuum chamber shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1, 5.3.3) to demonstrate the maintenance of mechanical properties.

5.4.4.5 Backing. Backing gas shall meet the composition requirements of 5.4.4.2. A change from backing gas to permanent or temporary backing, or vice versa, shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1, 5.3.3) to demonstrate the maintenance of mechanical properties. Backing material that is integrated into the weld, shall meet the requirements of 5.4.2.4. A change in backing material that does not interact with the weld and is a shield only, shall require a validation plate and supplemental PQR (see 5.5.3.2.1, 5.3.3).

5.4.4.6 Inert gas filled chamber. A chamber filled with inert gas may be used for shielding, without limit on the size of the chamber. The oxygen and nitrogen partial pressure shall be monitored. A change in the purging procedure shall require verification of the oxygen and nitrogen partial pressures. Change in shielding method from inert chamber to a trailing shield, backing gas, or combined flowed gas shall be cause for requalification. Change in shielding method from inert chamber to vacuum chamber shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1, 5.3.3) to demonstrate the maintenance of mechanical properties.

5.4.4.7 Vacuum chamber. A vacuum chamber may be used for shielding, without limit on the size of the chamber. The oxygen and nitrogen partial pressure shall be monitored. A change in the purging procedure shall require verification of the oxygen and nitrogen partial pressures. An increase in pressure in the vacuum chamber of greater than 20% of the originally

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qualified shall be cause for requalification. Change in shielding method from vacuum chamber to any other method shall be cause for requalification.

5.4.5 Joint design.

5.4.5.1 Joint design limitations . A change in any groove type to a square groove and vice versa shall be cause for requalification. A change beyond the fit-up tolerances listed in AWS D1.9/D1.9M “Limitations of Essential Variables of a WPS” table shall be cause for requalification

5.4.5.2 Joint soundness. If a joint design used in production inhibits accessibility or has repeated weld quality issues related to fusion, then it shall be changed to a more conservative design that allows for accessibility and ease of weld deposit.

5.4.5.3 Joint details for PJP . Any reduction in the root opening or decrease in bevel or groove angle from that qualified shall be cause for requalification. Any increase in depth of groove and resultant increased weld size from that qualified shall be cause for requalification as well. Therefore, any increase in the root opening and bevel or groove angles, along with any lesser joint thickness or depth of bevel below that qualified shall not be cause for requalification.

5.4.5.4 Backing or spacer strip. A change made from a weld made with backing to an open root weld excluding backing shall be cause for requalification. A change in the M number or class of the backing or spacer strip shall be cause for requalification.

5.4.5.4.1 Restrictions for using adhesive tape with ceramic backing materials. Unless used in initial qualification, the use of adhesive coated tape to hold ceramic backing material in place shall be prohibited. Ceramic backing tiles or fiber should only be used in conjunction with an inert gas purge. Adhesive materials shall be in contact with the workpiece only outside the heat affected zone.

5.4.6 Electrical parameters and technique.

5.4.6.1 Amperage. A change outside of the following limitations or a change that causes heat input to exceed the as qualified heat input limitation shall be cause for requalification:

a. For GMAW, an increase or decrease of 15%

b. For GTAW, an increase or decrease of 20%

5.4.6.2 Voltage. A change outside of the following limitations or a change that causes heat input to exceed the as qualified heat input limitation shall be cause for requalification:

a. For GMAW, an increase or decrease of 15%

b. For GTAW, an increase or decrease of 15%

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5.4.6.3 Travel speed. A change outside the following limitations or a change that causes heat input to exceed the as qualified heat input limitation shall be cause for requalification:

a. For GMAW, an increase or decrease of 15%

b. For GTAW, an increase or decrease of 15%

5.4.6.4 Wire feed speed. A change outside the following limitations shall be cause for requalification:

a. For GMAW, an increase of 20%

5.4.6.5 Heat input. Due to the effects of heat input on partial annealing, heat input shall be limited to the maximum value qualified in the PQR. A change in heat input outside the as-qualified maximum value qualified in the PQR shall be cause for requalification. The WPS electrical parameter ranges, and the high heat input values shall be limited to that which was qualified in the supporting PQR(s). Creation and implementation of a travel speed table should be listed in the entities WPS.

5.4.6.5.1 Bead width limitations. If maximum heat input is exceeded, it shall be a cause for requalification. Exceeding the maximum weave width qualified in accordance with 5.3.5 shall be cause for requalification.

5.4.6.5.2 Heat input calculations. Heat input shall be calculated. This may be performed by a. or b. for non-waveform controlled welding, or by b. or c. for waveform controlled welding.

a. Heat input [joules per inch] = $\frac{\text{Voltage x Amperage x 60}}{\text{Travel Speed [inches per minute]}}$

b. Heat input determined using instantaneous energy:

Heat input [joules per inch] = $\frac{\text{Energy [joules]}}{\text{Weld Bead Length [inches]}}$

c. For instantaneous power measurements in joules per second or watts:

Heat input [joules per inch] = $\frac{\text{Power [joules/second or watts] x Arc Time [seconds]}}{\text{Weld Bead Length [inches]}}$

5.4.6.6 Current type. A change from alternating current (AC) to direct current (DC) or vice versa shall be cause for requalification.

5.4.6.7 Polarity type. A change from direct current electrode negative (DCEN) to direct current electrode positive (DCEP) or vice versa shall be cause for requalification.

5.4.6.8 Output type. A change from Constant Current (CC) to Constant Voltage (CV) or vice versa shall be cause for requalification.

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5.4.6.9 Pulsed current. A change from pulsed current waveform to CC/CV waveform (non-pulsed) or vice versa shall be cause for requalification. A change from one pulsed current waveform to another shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1 and 5.3.3) to demonstrate the maintenance of soundness and mechanical properties.

5.4.6.10 Transfer mode. A change in GMAW transfer mode shall be cause for requalification.

5.4.6.11 Tungsten electrode. A change in electrode classification shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1 and 5.3.3) to demonstrate the maintenance of mechanical properties. A change in diameter outside the allowable range of AWS A5.12M/A5.12 shall be cause for requalification.

5.4.7 General procedure parameters.

5.4.7.1 Welding process. A change in welding process shall be cause for requalification.

5.4.7.1.1 Multiple welding processes used in PQR. A change in excess from what is allowed in 5.3.1.4 shall be cause for requalification. A root layer welded with any process not qualified to weld the root layer, as described in 5.3.1.4, shall be cause for requalification.

5.4.7.2 Welding method of application. A change in method of welding process application shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1 and 5.3.3) to demonstrate the maintenance of mechanical properties. A change between a manual or semiautomatic process to a mechanized, automatic, and robotic process application method shall be considered a change in the method of welding process application. See AWS A3.0M/A3.0 for definitions and “Method of Applying Welding Processes” Figure in AWS WHB-1.10 for further clarity of definitions.

5.4.7.2.1 Manual and semiautomatic applications. A change from a manual application to a semiautomatic or vice versa shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1 and 5.3.3) to demonstrate the maintenance of mechanical properties.

5.4.7.2.2 Mechanized, automatic, and robotic. Qualification using a mechanized, automatic, or robotic process application shall qualify the use of mechanized, automatic, and robotic process applications.

5.4.7.2.2.1 Adaptive control. Applications that use an adaptive control process shall be considered unique and shall each be qualified separately. The addition of, subtraction of, or change in adaptive controls shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1 and 5.3.3) to demonstrate the maintenance of mechanical properties.

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5.4.7.3 Position. A change in position exceeding the allowances of TABLE VII for Class I and the "WPS Qualification-Type of Weld and Position Limitations" table of AWS D1.9/D1.9M for all other classes shall be cause for requalification.

5.4.7.4 Preheat temperature. The addition or deletion of preheat, or increase of temperature by more than 18 °F (10 °C) shall be cause for requalification. This shall not include changes in ambient temperature within the allowances of 5.10.1. The WPS preheat temperature shall in no case be reduced below the as-qualified minimum PQR preheat temperature and shall be documented on the qualified WPS.

5.4.7.5 Interpass temperature. An increase in interpass temperature shall be cause for requalification. The minimum interpass temperature shall exceed the minimum preheat temperature. The maximum interpass temperature shall be documented on the qualified WPS.

5.4.7.6 Mulitpass and single pass welds. A change from a multiple pass weld to a single pass weld, or vice versa, shall be cause for requalification. A change outside the limits of the maximum single pass weld size, or the minimum multipass weld size shall be cause for requalification.

5.4.7.7 Backgouging. The exclusion of, but not inclusion of mechanical backgouging shall be a cause for requalification. A change made from a weld made including backgouging to an open root weld excluding backgouging shall be cause for requalification.

5.4.7.7.1 Method of backgouging. Mechanical backgouging shall be the only method of backgouging permitted.

5.4.7.8 Post Weld Heat Treatments (PWHT). The addition or deletion of PWHT, change of temperature by ± 18 °F (10 °C) or change in furnace atmosphere shall be cause for requalification.

5.4.7.9 Cleaning method. A change in cleaning method, type of chemical cleaning, sequence or parameters (see 5.11.3) shall require the preparation of a validation test plate and supplemental PQR (see 5.5.3.2.1 and 5.3.3) to demonstrate the maintenance of mechanical properties.

5.5 Welding Procedure Specification (WPS).

5.5.1 WPS requirements. Prior to any fabrication, a qualified PQR (see 5.3) shall be used to generate a welding procedure specification (WPS) in accordance with this document. The WPS shall be delivered to the procuring contract officer for approval upon request (see 6.2). The values listed on the WPS shall be determined by the allowable ranges in section 5.4 using the actual value recorded from the PQR as the reference point. Any change to an essential variable that falls outside the WPS ranges shall require a new PQR to create or to supplement a WPS. WPSs shall be signed and dated by the entity's responsible representative stating that the WPS and supporting documentation conform to the requirements of this standard.

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5.5.1.1 Supporting PQRs used for WPSs. A WPS may have more than one supporting PQR. One PQR or multiple PQRs may support more than one WPS. Supporting PQRs shall be based on weldability and compatibility to the original qualification intent and observe the limitation of each essential variable when growing the WPS ranges for applicability.

5.5.2 WPS essential variables. All WPS essential variables shall be in accordance with 5.5.1 and TABLE V. Any change to an essential variable in excess of the WPS ranges listed in 5.4 shall require requalification.

5.5.3 WPS transfer requirements.

5.5.3.1 Procedure transfer within an entity. Transfer of qualified procedures between sites, within an entity, shall be permissible providing the requirements stated in the entity's quality assurance manual and this document are met. The entity shall notify the PCO of the transfer prior to use at the entity's new site. Field service welding repair (see 3.10) activities shall be considered as the same entity, provided that both 1) personnel performing the work shall be from the same entity and 2) shall be guided by the same quality assurance manual as the original work was performed under.

5.5.3.2 Transferring of WPS from an entity to a sub-entity. A WPS for Class I welds shall only be transferred from an entity to their sub-entity under an active contract. The contracted sub-entity shall be known hence forth as "the transferee". The WPS and its supporting PQR(s) including all test results shall be provided to the transferee. The transferee shall pass a validation test (see 5.5.3.2.1) before using that WPS. The WPS, its supporting PQR(s), the validation PQR, and the test results shall be submitted to the procuring contract officer for approval prior to use (see 6.2). A WPS that is transferred shall meet the following requirements:

a. All required testing has been performed in accordance with TABLE VI, 5.7, and 5.9.

b. A successful PPIT has been performed in accordance with the requirements of Appendix A.

c. All the required nondestructive and destructive testing results are listed on the original PQR(s) and any applicable supplemental PQR(s) for the desired WPS to be transferred.

d. All essential variables of this standard are met and recorded in the original PQR, any applicable supplemental PQR(s), and the Class I WPS to be transferred.

NOTE: If the original PQR for the Class I WPS to be transferred is missing any of these requirements, a supplemental PQR shall be prepared in accordance with 5.3.3 by the original entity.

In no case shall the subcontractor extend the transfer of the Contractors WPS to another subcontractor. The test results and any transfer of a WPS shall be delivered to the PCO upon request.

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5.5.3.2.1 Validation test plate. The transferee shall prepare a validation test plate within the parameters recorded on the WPS to be transferred. The welded validation plate shall then be subjected to all of the required nondestructive (see 5.9) and destructive testing (see TABLE VI and 5.7), excluding the PPIT. All destructive test results shall meet the requirements of this document. All the validation testing results shall be listed on the validation WPS, which shall be differentiated from the original WPS transferred.

The validated WPS shall be certified by the transferee entity responsible representative. The validated WPS shall be permanently marked, including the addition of the validation WPS, and shall be uniquely identified for traceability. The validation WPS shall include its test results, along with the names, signatures, and dates of those involved in the validation testing process. The transferee validated WPS and all supporting documentation shall be delivered to the procuring contract officer upon request (see 5.5.3.2).

5.5.4 Contracted sub-entity qualification. It shall be the responsibility of the entity to ensure that their contracted sub-entities have their own qualified procedures that meet all the requirements of this document. For WPS transfer to a contracted sub-entity (see 5.5.3). A failure at the contracted sub-entity level for welding related issues that indicate significant quality concerns may include: Corrective Action Request (CAR) producing a Major or Critical Defect, systemic weld failures, or welding or welder negligence. These types of issues will require Government audits of the entity's welding quality system to determine conformance to this document or the contract. This could include but is not limited to: welding supplier quality audits, qualification approval process audits, or related inspection audits.

5.5.5 WPS revision. Any changes made to a WPS shall be documented on a revised WPS, then authorized and dated by the entity responsible representative. The entity shall notify the procuring contract officer that a change was made. The revised WPS and any supporting documentation shall be delivered to the PCO upon request (see 6.2).

5.5.6 WPS record retention. All WPSs shall be identified in a manner to allow permanent traceability. WPSs shall be retained as long as the WPS is applicable. Documentation of the supporting PQRs, and test results detailed in the welding procedure shall be kept by the entity as long as the WPS and the supporting PQR(s) are applicable. All WPSs, PQRs, and associated testing data shall be delivered to the procuring contract officer upon request.

5.6 Welder performance qualification.

5.6.1 Welder qualification requirements. Each entity shall establish that each welder and welding operator to be employed for welding has been qualified by demonstrating their ability to produce welds which meet the requirements of this document or AWS D1.9/D1.9M as applicable. The entity shall establish and maintain valid qualifications for each welder or welding operator prior to assigning any welder or welding operator to weld any contract deliverable components.

5.6.2 Welder Qualification Records (WQR). The entity shall document the results of the welder or welding operator qualification test in a WQR for every welder or welding operator. WQRs shall be delivered to the procuring contract officer for review and approval upon request.

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WQRs shall be maintained for the entirety of the welder or welding operator's term of employment, or period of performance of the contract, whichever is greater. A WQR example is shown in FIGURE B-3. The WQR shall include, at a minimum:

a. Entity name.

b. Welder or welding operator identification.

c. Date of test.

d. WPS used for WQR.

e. Essential variables listed in TABLE X.

f. NDT inspector information.

g. Results of applicable qualification test(s).

h. Entity responsible representative signature.

5.6.3 Welder and welding operator performance qualification. A qualified WPS (see 5.5) shall be used for each welder or welding operator qualification plate. Qualifying as a welder shall not qualify the individual as a welding operator or vice versa. A person tending a robotic work cell loading and unloading parts shall not require qualification. All welders and welding operators shall qualify on a groove weld prior to qualifying for plug, fillet, pipe or tube.

5.6.3.1 Ballistic welds. Prior to welding a ballistic weld, the welder or welding operator shall qualify in accordance with this document. All required qualification test plates shall be fabricated in accordance with the test plate assemblies shown in FIGURE 6 and FIGURE 7 and shall be complete joint penetration (CJP) welds. Qualification to FIGURE 6 shall qualify the welder or welding operator for 1/8 to 3/4 inch (3 to 19 mm) and qualification to FIGURE 7 shall qualify the welder or welding operator for 1/8 inch (3 mm) to unlimited thickness. The usage of backing and back gouging shall match the WPS. Essential variables for welder and welding operator qualification are defined in TABLE X. Qualification positions shall be in accordance with TABLE XI. Thicknesses qualified shall be in accordance with TABLE XII. All completed test plates shall pass a volumetric inspection (see 5.9) or destructive testing in accordance with TABLE XII. Qualification to an armor base material shall be considered as qualification to any armor or non-armor titanium alloy in accordance with this document.

5.6.3.2 External qualification for welding non-ballistic joints. If welding non-ballistic joints, the welder or welding operator qualification shall be in accordance with AWS D1.9/D1.9M or this document. Qualification to any non-armor titanium alloy shall be considered as qualification to any other non-armor alloy covered by this document. A welder or welding operator qualified to this document for ballistic welds shall also be qualified for welding non-ballistic joints in accordance with this document.

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5.6.4 Welder and welding operator essential variables. Each welder or welding operator qualification shall include the information from TABLE X for each qualification required. The applicable information from TABLE X shall be included in the WQR.

5.6.4.1 Welder and welding operator qualification positions. Each welder and welding operator shall be qualified in the position or positions (see TABLE XI) which shall be welded in accordance with the applicable WPS (see 5.5). Welder and welding operators who qualify in one or more positions shall be considered qualified in other positions in accordance with TABLE XI. For detailed descriptions of positions (see AWS A3.0M/A3.0).

5.6.5 Qualifying welder or welding operator through PQR. Any welder or welding operator that performs the welding of a PQR test plate, which passes all the qualification requirements, shall be qualified as a welder or welding operator in accordance with this document. All the applicable essential variables listed in TABLE X shall apply as limitations for this type of welder qualification and shall be recorded on the WQR.

5.6.6 Previous welder or welding operator qualification. Previous qualification tests by welders or welding operators that are properly documented and meet the requirements of 5.6.9 are acceptable (see 4.4) with the approval of the procuring contract officer (see 6.2).

5.6.7 Transfer of qualification. Performance qualifications shall not be transferred between entities. Transfer of performance qualification from one entity location to another location of the same entity is not permitted unless the welder's qualification meet the requirements of 5.6.3 and 5.6.9.

5.6.8 Renewal of qualification. The welder or welding operator shall requalify in accordance with this document for the following reasons:

a. Has not welded with the qualified process on titanium for a period exceeding six months.

b. A continuity record has not been maintained to provide traceability of process use.

c. As requested by entity or procuring contract officer due to a specific reason to question the abilities, performance, or quality of work of the welder or welding operator.

5.6.9 Maintenance of qualification. It shall be the responsibility of the entity to maintain a traceable continuity record for each welder or welding operator. Evidence of maintenance of qualification shall consist of at least one verification of process use for each qualified welding process during each six month period, and shall be provided to the procuring contract officer upon request. Additionally, verification of welding on titanium base metal must occur during a five year period, and shall be provided to the procuring contract officer upon request. If a continuity record cannot be produced when requested by the PCO or the entity, the welder or welding operator's qualification shall be considered expired and requalification shall be required. This continuity record shall be maintained through the entirety of each welder or welding operator's employment with the entity or the life of the contract, whichever is greater.

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5.6.10 Retest requirements. If a performance qualification test assembly fails to meet the applicable requirements, the initial test failure must be documented on the WQR. Retests may be allowed under the following conditions:

a. Two retest assemblies shall be welded without further training for each test assembly in each position which failed. Both test assemblies must be determined acceptable for the welder to qualify. Or

b. Subsequent retests may be welded for each test assembly that failed to meet the requirements of this document, if the welder or welding operator has had at least 4 hours of documented training or practice designed to correct the causes for previous failures. This training or practice and the initial test failure must be documented on the WQR.

5.7 Destructive testing of welds.

5.7.1 Destructive testing requirements. The type and number of destructive test specimens required for procedure qualification by both weld class and type are identified in TABLE VI. All destructive testing results shall be included in the PQR or WQR. Entities conducting destructive testing shall be ISO/IEC 17025 accredited, an AWS Accredited Testing Facility (ATF), a materials testing laboratory that is NADCAP accredited, or have an internal quality document for destructive testing. The internal quality document shall include procedures for all equipment used for testing and calibration records, documenting appropriate equipment calibration prior to use and in accordance with the applicable testing standard. If an internal quality assurance document is used, the document shall be delivered to the procuring contract officer upon request (see 6.2). Any entity that sub-contracts destructive testing shall ensure the test facility complies with this section.

5.7.1.1 Destructive testing personnel qualification. Entities that complete destructive testing internally shall develop a training plan for destructive testing personnel. The training plan shall be made available to the procuring contract officer upon request. Entities that sub-contract destructive testing shall use ISO/IEC 17025 certified test facilities, an AWS Accredited Testing Facility (ATF), or a materials testing laboratory that is NADCAP accredited. Visual inspection of bend, macroetch and PPIT samples must be completed by a certified visual inspector (see 5.8.2.1).

5.7.2 Tension/reduced-section tension testing. Tensile test specimens shall be extracted from the test plate as shown in FIGURE 4. Tensile test samples shall be prepared and tested in accordance with AWS B4.0.

5.7.2.1 Tension/reduced-section tension acceptance criteria. For acceptance of tensile test specimens, the tensile strength shall be greater than or equal to the specified tensile strength of the weakest material incorporated into the weld, according to MIL-DTL-46077 for armor grades or AWS D1.9/D1.9M Table “Strengths of Welded Titanium Alloys” for non-armor grades or as specified on the engineering drawing.

5.7.3 Bend testing. Bend test samples shall be prepared and tested in accordance with AWS D1.9/D1.9M “Bend Tests – Groove Welds – Plate and Pipe” for wrap-around guided bend

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specimens. All materials in accordance with MIL-DTL-46077 shall be held to a maximum bend radius of 18T. Samples shall be removed from the fixture when the outer roller has reached an angle of 120°. Number of wraparound guided bend tests are specified in TABLE VI. It is recommended that samples be a minimum of 8 inches (200 mm) long to accommodate the large bend radius of these materials. It is also recommended that corners radius be rounded to the maximum extent permitted in AWS D1.9/D1.9M. For test plate thickness (T) < 3/8 inch (10 mm), full thickness bend specimens should be used in place of side bend test specimens. Milling to reduced thickness is permitted down to 3/16 inch (5 mm). For plates ≥ 3/8 inch (10 mm), side bend specimens may be between 3/16 and 3/8 inches (5 mm and 10 mm). All specimens shall include the root. For grades with a high maximum radius requirement, thinner bend specimens are recommended. The total specified quantity shall be tested regardless of the type of bend test specimen.

5.7.3.1 Bend testing test acceptance criteria. For acceptance, test specimens after bend testing shall have no single discontinuity greater than 1/8 inches (3 mm). Cumulative discontinuity length shall not exceed 3/8 inches (10 mm) for all discontinuities greater than 1/32 inches (0.8 mm). Cracks occurring on the corners of the specimen shall be reviewed for evidence of inclusions or other fusion defects. If such evidence is found, the crack shall not exceed 1/8 inches (3 mm). Openings or failures in the base metal outside the weld material and heat affected zone shall not be cause for rejection.

5.7.4 Macroetch testing. Cross sections of weldments for macroetch testing shall be taken 1 inch (25.4 mm) or greater from the start or stop of any weld. The procedure for macroetch samples shall be documented and accompany the digital image. Macroetch of the weld cross section shall be recorded as a digital image with a reference scale embedded in the image. The digital image shall be attached to the PQR. See TABLE VI for number of specimens required. The quantity and location of specimens shall be taken equidistant about the weld length. Macroetch samples are required for all PQRs, in accordance with TABLE VI.

5.7.4.1 Macroetch acceptance criteria. For acceptance, the macroetch samples shall meet the visual inspection section criteria of MIL-STD-2035, class 2. Additionally, for acceptance, the weld metrics to define weld size shall not be less than the corresponding metric on the applicable fabrication document. If no weld size for PJP welds is specified in the applicable fabrication document, then the weld size shall not be less than the depth of the groove for acceptance. The macroetch samples shall meet the Macro Examination Acceptance Criteria of AWS D1.9.

5.7.5 Hardness testing. When required by the procuring activity, hardness testing shall be in accordance with ASTM E10, ASTM E18 or ASTM E92.

5.7.6 Proofing projectile impact testing. All PPIT for ballistic procedure qualifications shall be performed in accordance with TOP-2-2-711.

5.7.7 Destructive test failures. If a single destructive test specimen of each test type from a qualification plate fails to meet the acceptance criteria, a retest of that plate may be performed. The retest shall include two additional specimens from the same qualification plate for the failed destructive test. The retest shall only be accepted if both additional specimens meet the acceptance criteria. If the original qualification plate lacks material for additional required

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test specimen, then a supplemental test plate shall be created using the original parameters, to support additional specimen generation. If any additional test specimen fails to meet the acceptance criteria, that qualification attempt shall be deemed unacceptable, no retesting shall be allowed, and a new qualification test plate shall be fabricated.

5.8 Nondestructive Testing (NDT) personnel qualification.

5.8.1 Requirements. Individuals performing any method of NDT shall meet the qualification requirements for the applicable NDT level of certification in accordance with ANSI/ASNT CP-189 or AIA/NAS410, except as listed below. The required NDT level shall be determined by the contractor's NDT written practice (see 5.9.1.1). Alternately, any certification through ASNT Central Certification Program (ACCP) shall be recognized as certified at the NDT level of examination. The certified inspector shall determine compliance of the weld to the requirements of this standard (see 5.9.1). When the results of an inspection process or indications are inconclusive, the weld shall be inspected by another NDT method.

5.8.1.1 Entity responsibility. It is the responsibility of the entity to ensure that all subcontractors are qualified and certified to conduct all nondestructive testing in accordance with this standard.

5.8.2 Visual inspector qualification. Certified visual inspectors (CVI) and authorized visual inspectors (AVI) shall be used for the verification of weld quality. All visual inspectors shall undergo a near vision eye exam which shall be administered every three years, including a color perception test, unless otherwise specified by the certifying authority for certified visual inspectors. The CVI eye exam shall be administered by a qualified, licensed medical professional, while AVI eye exams may be performed by the entity.

NOTE: A visual inspector is not qualified to perform any other type of NDT unless the requirements of 5.8.3 are met.

5.8.2.1 Certified visual inspectors. Certified visual inspectors (CVI) shall have the authority to sign as certified inspectors in accordance with this document and shall be certified in accordance with at least one of the following:

a. Current certification in accordance with the AWS QC1 for Certified Welding Inspector (CWI) or Senior Certified Welding Inspector (SCWI).

b. Current certified welding inspectors qualified by the Canadian Welding Bureau (CWB) to Level II or the Level III requirements in accordance with CSA W178.2.

c. Current Level II Visual Inspection (VT) certification or greater in accordance with the ANSI/ASNT CP-189.

5.8.2.2 Authorized visual inspectors (AVI). Welding inspectors qualified in accordance with this method herein may visually examine welds while working under active supervision (see 3.1) with the certified visual inspector, but shall not have the authority to sign as a certified visual inspector on welds. Weld examination approval for visual examination shall be the

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responsibility of a Certified Visual Inspector, as listed in 5.8.2.1. Visual inspectors qualified to documents other than what is listed in 5.8.2.1 must be trained and qualified by a visual inspector training program that shall be in accordance with AWS B5.1, AWS B5.2, ASNT SNT-TC-1A, or an equivalent document and authorized by the Government procuring contract officer Certified Visual Inspector (see 5.8.2.1).

5.8.2.2.1 Previous CVI certifications. Previous certification in accordance with the 5.8.2.1 precludes the need for the additional training specified in 5.8.2.2 and allows for immediate qualification as an AVI. Those qualified in this manner shall be restricted to within two years following the expiration of their certificates. In cases where the external qualification has lapsed, then the vision test must be passed before being able to perform as an AVI.

5.8.3 Other NDT methods. All NDT personnel shall be qualified and certified in accordance with ANSI/ASNT CP-189 or AIA/NAS410. For NDT methods not covered by ANSI/ASNT CP-189 or AIA/NAS410, personnel shall qualify to comparable levels of competency by the administration of examinations for the particular method involved. A certified NDT Level III inspector shall be responsible for the validity of the examinations and overall personnel qualification and certification (PQ&C) program.

a. Liquid Penetrant Testing (PT)

b. Radiography Testing (RT)

c. Digital Radiography Testing (DR)

d. Radiograph Interpreter

5.8.4 NDT re-examinations. If the Government procuring contract officer NDT or welding engineering representative has reason to believe that an individual is unable to competently perform at the NDT level that the individual is certified or because of interrupted service the entity shall request a practical or written examination be administered in accordance with ANSI/ASNT CP-189 or AIA/NAS410. If an individual transfers from one entity to another, NDT reexamination shall be required in accordance with the new entity's NDT written practices for personnel qualification and certification. All personnel certified through a nationally recognized certification program such as CWI, CWB, or ASNT, shall be exempt from reexamination when transferring from one entity to another.

5.8.5 Alternative NDT specifications and standards. If an alternative inspection program which meets similar requirements of this document is to be used, the alternative program shall be submitted to the procuring contract officer for approval by an applicable Government NDT Level III prior to being used (see 6.2).

5.8.6 Personnel qualification documentation. All personnel qualification documentation shall be maintained for the entire length of employment of the individual or the life of the contract, whichever is greater. The procuring contract officer may conduct reviews of the personnel documentation to assess conformance to this standard at any time.

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5.9 Nondestructive testing (NDT).

5.9.1 Nondestructive Testing (NDT) requirements. This section covers the requirements for conducting NDT used in determining the presence of discontinuities. Unless otherwise specified, all NDT shall be performed in accordance with the requirements of this section. This document shall serve as the fabrication document, when a fabrication document is referenced in any other document, unless otherwise required by the technical authority. All classes of welds shall require some form of NDT: see TABLE XIII for inspection method type as required per class of weld. The method of NDT shall be specified in the contract or drawing. When NDT has an indication of a flaw, the process of evaluation shall be in accordance with FIGURE 8.

NOTE: The terms inspection, testing, and examination are commonly used interchangeably in NDT.

5.9.1.1 NDT written instructions. When NDT is performed, written instructions for performing the examination shall be prepared and uniquely identified. The written instructions shall have written certification from an applicable NDT Level III inspector (see 5.8.3). The procuring contract officer may conduct reviews of the NDT written instructions to assess conformance to this standard. All NDT procedures shall be performed in accordance with the NDT written instructions. Typical NDT written instruction should include (but not limited to):

a. Scope, purpose, application.

b. Equipment involved, specifications for equipment, and consumables.

c. Test setup qualification and certification of calibration shall be referenced.

d. Required qualifications of personnel required to perform to said instruction.

e. Reference to external test method procedures, for example ASTM.

f. Details of to which, what, and how the instruction is applied and performed.

g. Acceptance criteria.

h. Recording and reporting results

5.9.1.2 Surface preparation. Weld contours shall not interfere with the NDT process or interpretation of the NDT results. As determined by the responsible Level III NDT inspector, NDT written instructions, or applicable standard, excessive surface irregularities and tightly adhering spatter shall be removed by a mechanical process, such that they will not interfere with the interpretation of the nondestructive test results. All non-permanent backing material shall be removed prior to the performance of any NDT (see 5.1.5).

5.9.1.3 Frequency of NDT examination. All welds shall be visually inspected at a 100% frequency without deviation. Welds to be covered shall be inspected prior to being made

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inaccessible (see 5.9.2.1.2). A continuous sampling plan shall be developed for all NDT activities in accordance with MIL-STD-1916. The verification level (VL) for ballistic welds shall be VL V. If an alternate acceptance method is requested, the requirements and procedure shall be in accordance with MIL-STD-1916.

5.9.1.4 Inadvertent examination. Any indication that is discovered through inadvertent examination shall follow the NDT evaluation process (see FIGURE 8). Any indication that is interpreted as relevant shall be evaluated for acceptability. If the indication is evaluated as a reject, the defect shall be repaired in accordance with 5.12, and shall be examined after final repair with the same NDT method of the original inadvertent inspection. If the initial examination was performed with RT or UT and the repair does not exceed two layers or half the thickness of the thinner member, whichever is less, then VT is acceptable for examination after the repair. Any repairs shall be recorded in the NDT examination report.

5.9.1.4.1 Inadvertent defects. When an inadvertent examination discovers a rejectable indication in a particular joint, the remainder of that joint shall be examined. Any additional indications shall be interpreted in accordance with 5.9.1.4

5.9.1.5 NDT examination reports. When NDT is performed, an NDT examination report shall be prepared. The NDT examination report shall be made available to the procuring contract officer upon request. The NDT examination reports shall include, at a minimum, the following:

a. Method of examination.

b. NDT written instruction unique identification (see 5.9.1.1).

c. Date and time of inspection.

d. Acceptance criteria used.

e. Weld joint design, classification and unique identification.

f. Welding procedure specification (WPS) identification.

g. Stage of manufacturing (initial inspection, in process, final, or repair).

h. Test results (accept or reject).

i. Signatures of responsible NDT level II or III inspector.

5.9.1.6 NDT symbols. When NDT symbols are used on drawings, the symbols shall be in accordance with AWS A2.4.

5.9.2 Nondestructive Testing (NDT) methods.

5.9.2.1 Visual testing examination (VT). All completed welds shall be visually examined by a visual inspector (see 5.8.2). No unspecified welds shall be added without

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approval of the procuring contract officer. The visual testing procedure shall contain, as a minimum, the requirements of 5.9.1.5 and the following information:

a. Type of welds or surfaces to be inspected.

b. Specific measuring devices to be used.

c. Visual aids, reference standards, workmanship standards.

d. List of inspection attributes (visual characteristics).

e. Lighting requirements.

f. Acceptance criteria and classification of defects.

g. Record requirements.

5.9.2.1.1 Visual testing surface illuminance. The surface illuminance shall be 540 lux (lx), or 50 foot-candle (fc), at a minimum. The illuminance source shall be held at a minimum of 6 inches (152 mm) from the piece to be inspected and a maximum distance of 18 inches (457 mm).

5.9.2.1.2 Visual testing examination access. For direct examination, access to the weld shall be such that the weld inspector's eye shall be within 10-24 inches (254-610 mm) of the examined surface. The angle of examination shall not be less than 30 degrees from the surface of examination (see FIGURE 9). The usage of cameras, fiber optic cables, mirrors, and borescopes shall be acceptable for visual testing when access in accordance with AWS B1.11M/B1.11 is not possible or when required by the applicable standard or drawing.

5.9.2.1.3 Visual testing acceptance criteria. The acceptance criteria for all welds shall be in accordance with TABLE XIV and 5.9.2.1.3.1, and shall meet the requirements of the applicable fabrication document or material specification. Weld profile, weld reinforcement, and undersized weld limitations shall be in accordance with AWS D1.9/D1.9M "Visual and Penetrant Acceptance Criteria".

5.9.2.1.3.1 Weld Appearance. Prior to altering the as-welded surface, the welder or certified visual inspector shall ensure the criteria of this paragraph are met. All weld surfaces and surrounding 1/32 in. (0.8 mm) of material shall exhibit a bright silver or silver luster. Other conditions shall be addressed as follows:

a. A first occurrence of straw, bronze or brown color shall be cleaned prior to proceeding. Shielding should be verified and corrected.

b. A third occurrence of straw, bronze or brown on the initial qualification weldment, or on a replacement after a test failure, shall be cause for rejection of that weldment.

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c. Loss of luster or other colors shall be cause for rejection and requires shielding deficiencies as well as welding deficiencies be addressed.

5.9.2.1.4 Visual testing reports. A visual testing report documenting the test results shall be developed for all VT and shall be made available to the procuring contract officer upon request. If a discontinuity is found that requires repair, full traceability and detailed records of all VT activities shall be undertaken and delivered to the PCO upon request. The entity's CVI or AVI shall make certain only applicable WPSs that are qualified and approved per this document are used and have been documented accordingly. Records of visual inspection shall contain the requirements of 5.9.1.5 as a minimum.

5.9.2.2 Liquid penetrant testing (PT). When required, liquid penetrant testing (PT) shall be used for detecting the presence of surface discontinuities in all classes of weld. PT shall be performed in accordance with ASTM E165/E165M and ASTM E1417/E1417M for all class welds. If rework is required following (PT), cleaning methods (see 5.11.3) and preheat (see 5.10.4) must remove all traces of liquid prior to welding. Due to the likelihood of contamination, this method is not recommended unless other inspection methods are not available for use. The vaporization temperature of the penetrant selected shall not exceed the preheat and interpass temperature limit of the base material.

5.9.2.2.1 Liquid penetrant testing surface preparation. All surfaces to be examined shall be prepared in accordance with ASTM E1417/E1417M.

5.9.2.2.2 Liquid penetrant testing acceptance criteria. The acceptance criteria for the welds shall be in accordance with TABLE XIV.

5.9.2.2.3 Liquid penetrant classification. The penetrant examination process and materials shall be classified in accordance with ASTM E1417/E1417M.

5.9.2.2.4 Liquid penetrant testing reports. At a minimum, reports shall contain the criteria outlined in 5.9.1.5.

5.9.2.3 Radiographic testing (RT). When required, radiography by X-ray or gamma ray sources shall be used for determining the presence of discontinuities. RT shall be performed in accordance with ASTM E1742/E1742M and ASTM E94/E94M or ASTM E1032.

5.9.2.3.1 Radiation imaging systems. Ionizing radiation methods other than RT may be used to perform the examination of welds. When a radiation imaging system is used to perform the examinations, the procedures and techniques shall be in accordance with AWS D1.9/D1.9M, Radiographic Procedures section.

5.9.2.3.2 Radiographic testing acceptance criteria. Acceptance criteria for RT shall be in accordance with TABLE XV.

5.9.2.3.3 Digital radiography acceptance criteria. The acceptance criteria for DR shall be in accordance with the TABLE XV.

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5.9.2.3.4 Radiography and digital imaging reports. RT and DR inspection reports shall contain, at a minimum, the requirements of 5.9.1.5 and shall be in accordance with the applicable standard or technical bulletin.

5.9.2.3.5 Radiographic film and digital imaging storage. All radiographic film shall be stored in accordance with ASTM E1254. All digital radiography data files shall be in accordance with ASTM E2699.

5.9.2.4 Other NDT methods. If NDT methods, not covered in ANSI/ASNT CP-189, or AIA/NAS410, which meet all the requirements of this document are to be used, the alternative procedure and written practices shall be submitted to the procuring contract officer upon request and reviewed by the applicable Government NDT Level III for approval prior to being used (see 6.2).

5.9.2.4.1 NDT technical manuals. The Tri-services technical manuals (TMs) for NDT, TM 1-1500-366-23 and TM 1-1500-335-23, may be used as a replacement for the requirements in 5.9.2. If the TMs are used, the acceptance criteria shall be agreed upon in the contract documents prior to performing NDT. If the acceptance criteria are not provided in the contract, then the acceptance criteria of this standard shall be used for each NDT method. If a new TM is to be developed as NDT written instructions (5.9.1.1), then MIL-HDBK-1222 should be used as a guide.

5.9.3 Nondestructive testing record retention. All NDT examination records shall be retained for the period of performance of the contract, at a minimum. NDT records shall be delivered to the procuring contract officer upon request.

5.10 Welding requirements.

5.10.1 Welding environment. Barriers for protection from wind and inclement weather shall be established prior to any welding and shall be maintained until the completed weld has cooled to within 50 °F (10 °C) of the ambient air temperature. Environment temperature differentials of greater than 100 °F (20 °C) along the weld joint during welding and cyclic heating should be avoided to minimize stress differentials within the completed weld.

5.10.1.1 Wind. The maximum wind velocity in the near vicinity of the welding process shall be no greater than 5 miles per hour (8 kilometers per hour).

5.10.1.2 Air temperature. The air temperature in the nearby vicinity shall not be lower than 60 °F (16 °C).

5.10.1.3 Precipitation. Welding shall not be performed when material surfaces are exposed to any form of precipitation or if the material surfaces are wet.

5.10.1.4 Base material temperature. Welding shall not be performed when the temperature of the weld joint is below ambient air temperature, nor when material temperature is below 60 °F (16 °C). If the material or ambient temperature is below 60 °F (16 °C), welding shall not be performed until the material is brought above 60 °F (16 °C).

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5.10.2 Weld Procedure Specification (WPS) accessibility. It shall be the entity's responsibility to ensure that the WPS is readily accessible to welders and welding operators during fabrication. The entity shall have an established system to ensure the accurate application of the parameters of the applicable WPS to any fabrication done in accordance with this document. This system shall be verified by the entity's welding engineer or quality assurance personnel and defined in an entity's quality management system procedure.

5.10.3 Position. Welding positions shall be in accordance with the qualified position of the applicable WPS. See AWS A3.0M/A3.0 for position descriptions.

5.10.4 Preheat and interpass temperatures. Preheat and interpass temperature monitoring is required and shall be in accordance with the WPS. If these temperatures are exceeded, the base material shall be repaired or replaced, see 5.12 for repair limitations.

5.10.4.1 Preheating and interpass temperature measurement. Preheat temperature shall be checked prior to welding as described in 5.10.4.1.1. Interpass temperature shall be checked prior to initiating another weld pass as described in 5.10.4.1.2. Temperatures of 125 °F and above shall be checked using temperature measuring devices. The use of temperature measuring crayons is prohibited. If there is evidence of moisture on the material surface, preheat shall be applied until the area is dry. All temperature measuring equipment shall be calibrated or verified to a calibrated meter (see 4.9.2). Entities that wish to use procedures utilizing prohibited measurement tools shall deliver these procedures for approval by the procuring contract officer upon request (see 6.2).

5.10.4.1.1 Preheat temperature measurement. Preheat temperature shall be measured on the surface of the base material on the side from which welding will be performed and 3 inches (76 mm) in all directions from the area to be welded along the joint.

5.10.4.1.2 Interpass temperature measurement. Interpass temperature shall be measured at the edge of the joint on the surface of the base material on the side from which welding was performed, as close as feasible to the start of the next weld pass, or continuation of a weld bead if the weld is not momentarily reinitiated.

5.10.4.2 Heating. The heating shall be of the uniform soaking type and uniformly distributed on and around the area being welded. Spot heating shall be prohibited. The spacing and wattage of heaters shall be such as to ensure that the entire welding area is maintained within the required temperature range. Heat should be applied by any of the following suggested methods used either individually or in combinations:

a. Resistance heaters.

b. Radiant and infrared heaters.

c. Electrical induction.

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5.10.4.3 Control of minimum temperature. The minimum required temperature shall be established prior to welding. Loss of the minimum required temperature shall require reheating, as necessary, prior to resuming welding. Reheating to the specified minimum interpass temperature range prior to the resumption of welding shall be done before or after this inspection.

5.10.4.4 Control of maximum temperature. The maximum required temperature shall be checked as specified in 5.10.4.1.2. If at any time during welding (except during deposition of a weld pass) the base metal temperature is found to be above the maximum interpass temperatures noted in the WPS, the weld area shall be allowed to cool to within the required preheat temperature prior to any subsequent welding.

5.10.5 Post Weld Heat Treatment (PWHT). PWHT shall require PCO and TA approval. Recommendations for specific procedures are available in AWS WHB-5.9 Annealing and Stress Relief Heat Treatments for Titanium Alloys table.

5.10.6 Tack welding. If tack welding is used as an aid in fabrication, the welding shall be performed by a qualified welder or welding operator. All tack welds shall be in conformance with the appropriate WPS, preheat and interpass requirements, and surface color requirements (see 5.9.2.1.3.1) that apply to final welds. Tack welds should be of sufficient length or size to overcome strain issues and be inspected for cracks. Cracked tack welds and those of poor quality shall be removed with backgouging or mechanical method as appropriate. For unincorporated tack welds (see 5.10.7).

5.10.6.1 Incorporated tack welds. Incorporated tack welds shall be considered initial increments of the final weld and meet the applicable VT acceptance criteria prior to incorporation. Welds shall be properly prepared for subsequent weld tie-in or additional passes. Incorporated tack welds shall be dressed and blended with minimal geometry and favorable profile before being incorporated with subsequent welding. Alternative conditions for remelting of tack welds shall be directed to the TA for review.

5.10.7 Temporary welds for fabrication purposes. Temporary attachments, temporary welds, and unincorporated tack welds used as construction aids to facilitate handling, positioning, and distortion mitigation are allowed, but should be used minimally. The same WPS, quality, welding, and workmanship requirements of this document shall apply to temporary attachment welding and removal. Construction aid material shall be compatible with the base metal that is being joined. Sufficient preheat, attention to heat input, adequate inert gas shielding and avoidance of damage to the base material during temporary welding and removal is required in accordance with this document. Attachments should be removed a minimum 1/8 inch (3 mm) away from the weld, followed by careful grinding, and blended smooth and flush into the adjacent base material surface. Attachment removal locations shall be documented on an inspection report. The removal area is required to pass a successful VT inspection after attachment removal and any subsequent allowable base material repairs to be considered acceptable (see 5.9 and 5.12).

5.10.8 Weld profiles. All weld profiles shall meet the requirements of AWS D1.9/D1.9M “Acceptable and Unactable Weld Profiles”.

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5.10.9 Arc strikes. Arc strike inspection and limitations shall be in accordance with TABLE XIV. For removal and repair of arc strikes (see 5.12).

5.10.10 Weave and stringer passes. Weaving and stringer passes may be used interchangeably, so long as the heat input (see 5.4.6.5) and bead width qualification limitations (see 5.3.5 and 5.4.6.5.1) of this document are met, and the essential variable ranges of the qualified WPS (see TABLE V and 5.4) are not violated. Note, the use and need of both stringer beads and weaves beads in a single PQR or WQR qualification test assembly is required, just the same as is practical to facilitate and promote soundness in different joint designs and joint geometry encountered in production.

5.11 Workmanship requirements.

5.11.1 Weld joint surface preparation. All paint or coatings that would contaminate welding shall be removed for a minimum distance of 3 inches (76 mm) in all directions from the joint to be welded. Surfaces to be welded upon and adjacent surfaces at a minimum distance of 1/2 inch (13 mm) from the expected weld area shall be clean, dry and free of surface matter and defects such as:

a. Paint, except in accordance with the 3 inch (76 mm) minimum requirement stated above.

b. Oil, grease.

c. Moisture.

d. Mill scale.

e. Objectionable nicks, gouges and irregularities.

f. Excess slag.

5.11.2 Pre- and in-process weld cleaning. The weld joint and surrounding area shall be cleaned to remove surface contaminants and the titanium oxide layer. The cleanliness of this area shall be maintained prior to and during welding. A multistep cleaning procedure is recommended as each step is optimized at removing a contaminant. The cleaning procedure must remove sources of oxygen, hydrogen and any aggregates from the surface to be welded. Detergent wash and solvent wash or wipes that evaporate are considered insufficient to remove the oxide layer. Cleaning methods are detailed in 5.11.3. of this document. A surface that has been cleaned and protected from ambient contaminants shall be considered clean for a period not to exceed 24 hours unless isolated from oxygen via vacuum or inert gas. After cleaning and prior to welding, the area shall be dry and free of surface matter. Cleaning methods involving chlorine (such as bleach) or chlorinated solvents such as trichlorethylene or methylchloride are prohibited.

5.11.3 Weld cleaning methods. Cleaning methods shall be labeled according to method utilized and objective in alignment with TT-C-490. Methods of surface cleaning and other

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specific variables are listed in TABLE XVI. The cleaning method utilized and all critical variables shall be identified and recorded on the PQR. This shall include method, materials and time as required in the methods identified below. Upon acceptance of the PQR, cleaning methods and variables shall be documented on the WPS. Other methods not prohibited by drawing or contract may be used, provided the requirements in this specification are met. Failure to identify the cleaning method and control variables is cause for rejection of the PQR and associated WPS. Cleaning methods include:

a. Method 1 – surface removal. This includes mechanical wire wheel, mechanical milling, bead shot blast, abrasive blast, sanding, and grinding. Aggregate particles from abrasive blast, sanding, and grinding must be removed by a subsequent fluid cleaning method. Wire wheel, dry milling and bead shot blast may be performed at elevated temperature. If cutting fluid is used, a subsequent wash method is required. Debris must be removed prior to the resumption of welding.

b. Method 2 – solvent cleaning. This method addresses surface contaminants only. This includes immersion, spray, vapor and hand wiping with solvent fluids.

c. Method 3 – detergent cleaning. This method addresses surface contaminants only. This includes spray, hand wiping, immersion, ultrasonic, and hot alkaline methods.

d. Method 4 – emulsion cleaning. This method addresses surface contaminants only. This includes emulsions with and without water. A subsequent fluid cleaning or water rinse is recommended.

e. Method 5 – deoxidation. This method addresses surface oxides. This includes acid spray, base spray, other spray, acid immersion, base immersion, other immersion, chemically assisted ultrasonic, and hot alkaline electrolytic methods. These methods should be followed by a rinsing operation.

f. Method 6 – non-oxide chemical conversion of the surface. This method is not applicable to titanium.

g. Method 7 – steam cleaning. This method addresses contaminants only. This includes steam cleaning with or without pressure washing assistance.

h. Method 8 – laser cleaning. This method addresses surface contaminants only at low power, and both contaminants and surface oxides at high powers. This method can be used at elevated temperatures when coupled with vacuum or in-process blown gas. Room temperature may be coupled with vacuum or in process blown gas or followed with a second fluid method to address debris produced by the process. MIL-HDBK-529 provides guidance on best practices.

5.11.3.1 Weld filler metal cleaning. Cleanliness of the material shall be audited to ensure that it arrives at the point of use clean in accordance with the original filler material requirements, documented by the entity’s quality management system. If this chain is broken, then the filler metal shall be cleaned using a validated procedure. Filler storage requirements shall be in accordance with 5.2.2.2.2.

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5.11.3.2 Interpass cleaning. Cleaning between passes may be used to limit the oxygen buildup in the weld and shall be documented in the WPS. If used, the cleaning procedure shall be documented in accordance with TABLE XVI, and is considered an essential variable (see 5.4.7.9 and TABLE V). Interpass cleaning can be completed while maintaining temperature or performed after cooling to room temperature. Methods requiring subsequent fluid cleaning steps shall not be used for elevated temperature cleaning, but dry machining is allowed. Ambient temperature cleaning may include fluid cleaning steps, provided the weld is thoroughly dried prior to the continuation of welding. If the room temperature interpass cleaning is due to a pause in the welding, but is not a requirement on the WPS, then one of the liquid removal processes may be used to remove dust, lint, oils, etc. prior to the resumption of welding.

5.11.4 In-process loss of shielding. A loss of shielding in process shall require visual inspection prior to resumption of welding (see 5.9.2.1.3.1). If the portion of the bead generated without proper shielding is unavailable for visual inspection, all weld material shall be removed prior to the resumption of welding. In cases where the point of shielding loss is unknown, all weld material shall be removed prior to the resumption of welding.

5.11.5 Completed welds. All tightly adhering weld spatter greater than 1/8-inch (3 mm) diameter and residues shall be removed. Weld spatter that cannot be removed by standard cleaning methods may remain unless there are functional or cosmetic concerns or the spatter would interfere with NDT (see 5.9.1.2). The method or combination of methods used shall be selected to suit the properties of the metal, the type of contaminates and the degree of contamination present.

5.11.6 Welder and welding operator workmanship check. Prior to and during welding, welders and welding operators shall ensure base metal is free of foreign materials and discontinuities that would inhibit successful welding. All weld joints shall receive a baseline visual check by the welder or welding operator prior to, during, and after welding, and prior to involving a NDT inspector. Any concern observed by the welder or welding operator shall be reported to a CVI or AVI for VT inspection. The welder or welding operator does not need to be a qualified weld inspector to perform this quality check. Regardless of qualifications, the welder or welding operator shall not have the ability to approve the final inspection on their own work.

5.11.7 Joint fit-up inspection. Joint preparation shall be in accordance with approved configurations including those in approved welding procedures. When fit-up tolerances exceed the limitation of approved joint designs, correctional buildup by welding may be employed within the limitations of 5.12.4.2. For fillet weld joints, when the fit-up gap exceeds 1/16 inch (1.6 mm) as a nominal condition along the joint, fillet weld size shall be increased by an amount equal to the excess of opening above 1/16-inch (1.6 mm). Where the gap between members exceeds 3/16-inch (4.7 mm) as a nominal condition along the joint, the condition shall be corrected by weld buildup in accordance with 5.12.4.2 of this document or the joint shall be prepared as a full penetration weld. Weld buildup shall not be performed on armor material.

5.11.8 Distortion control. All efforts to mitigate distortion shall be made on all weldments. Some general practices to mitigate distortion are as follows:

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a. Alternating weld deposit sides to keep a balance of weld deposit on the front and back side of a weldment.

b. The use of backstep sequences.

c. Depositing weld from the middle of the weld joint and working toward the edges of the work piece. Weld starts should be removed and blended to avoid discontinuities.

d. Alternating weld deposit directions.

e. Using larger stringer beads, as opposed to smaller beads, to slow the cooling rate.

f. The use of strongbacks, fixtures, construction aids, etc. to brace the weldment. Allow the weldment to cool to ambient temperature prior to removal of these apparatuses.

g. Pre-cantering or offsetting the weld joint by its anticipated distortion.

h. Limit oversized welds.

i. Using stitching to limit heat input into the weld joint.

5.12 Repairs.

5.12.1 Repair requirements. This standard is not intended for specific repairs of individual systems. Repairs to a system are performed after manufacturing has been completed and accepted by procuring contract officer. All repair procedures shall utilize a WPS developed and qualified in accordance with this document. Entities that wish to use proprietary repair procedures shall deliver these procedures for approval by the procuring contract officer upon request (see 6.2). Before the repairing of any component can be accomplished, the base material shall be prepared in accordance with 5.11.1. Repairs 5/8 inches (16 mm) or greater from the weld toe line shall be considered outside the weld zone and considered base material repairs. All materials fabricated out of tolerance shall be repaired until all tolerances and requirements are met. All repairs shall be inspected with the applicable NDT process specified for the weld class (see TABLE XIII).

5.12.2 Repair procedures. Repair procedures shall be developed and performed in accordance with this document. System technical bulletins, technical manuals, applicable military standards, or military specifications may not be used unless specifically approved by the procuring contract officer. All repair procedures shall be delivered to the PCO for approval upon request. After repair procedures are completed, the fabricated material shall meet all requirements listed in this document, applicable contracts, standards, or drawings.

5.12.3 Repair cycles. Repair cycles shall be referred to and noted on inspection or repair reports as R1 (one repair performed) or R2 (two repairs performed) or more if needed. Minor repairs such as undercut, overlap, and visual blends, shall not be considered in the previous context. Only major defects such as incomplete fusion and cracks or other defects that have failed volumetric NDT and require significant excavation and re-welding shall need

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documentation. The Government reserves the right to review any or all MRB processes, procedures, and documentation.

MRB disposition shall be required when any one of the following conditions exists:

a. When more than two weld repair attempts have been performed at the same location on armor.

b. When more than three weld repair attempts have been performed at the same location on materials not listed in a).

c. When the wrong filler metal has been used.

d. When a repair weld is required after the weldment has been postweld heat treated.

e. When a repair weld is required after final machining has been completed.

f. When the repair extends beyond the original weld zone (fusion zone and HAZ) and any additional removal requirements of this document.

g. When a weldment has been made with parameters outside the qualified WPS range.

h. Any plug, slot, or hole repairs.

i. Weld repairs following proof or leak test.

j. Any hole repairs on armor material or holes outside of allowances.

k. Weld buildup on armor material or outside the allowances of 5.12.4.2.

5.12.4 Repair of armor base material. Authorized repairs of armor base materials require a qualified WPS in accordance with 5.3.6.1 and a detailed repair procedure proposal. This includes repairs of mill defects. This repair procedure proposal shall state at a minimum: the extent of the repair, its location, depth, and size, the impact it will have on the mechanical and ballistic properties, and any required qualification documentation to achieve a sound repair, including a volumetric inspection plan.

5.12.4.1 Arc strikes, nicks, gouges, and other fabrication scars. Arc strikes, nicks, gouges, and other fabrication scars up to and including 1/32 inch (0.8 mm) in depth are acceptable without repair. Arc strikes, nicks, gouges and other fabrication scars in excess of 1/32-inch (0.8 mm) in depth, but within the limitations listed shall blend smoothly into the base material or shall be repaired by mechanical means. Arc strikes, nicks, gouges and other fabrication scars in excess of the below listed shall be repaired by welding. For a base material thickness less than ½-inch (13 mm), the maximum allowable depth is 1/32 inch (0.8 mm). For base material thickness greater than ½-inch (13 mm), a depth up to and including 1/16-inch (1.6 mm) is allowed. The final repaired surface shall blend smoothly into the base material and shall not interfere with clearance.

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5.12.4.2 Weld buildup. Buildup around the weld joint shall not exceed a total of ½ T of the thinner member or ½-inch (13 mm), whichever is less. Filler material must meet or exceed the base material specified minimum tensile strength. The buildup shall be considered part of the weld involved and shall be inspected in accordance with its specified classification.

5.12.5 Weld metal removal. When a defect in the weld metal is discovered during inspection or testing, the defect shall be removed. Weld metal containing the defect shall be completely removed and then prepared with a groove angle in accordance with applicable WPS or 40° minimum, and the end of the weld metal junction shall be tapered to an approximate 45° slope (see FIGURE 10) to allow for ease of reentry. The prepared repair excavation length, where permissible by material geometry, shall be the original defect length plus 4t, where t is the prepared repair depth thickness. The depth of the removed weld metal shall be the depth of the defect plus an additional 1/8 inches (3 mm) where permissible by material thickness. Once weld metal has been removed, a visual inspection and appropriate NDT shall be performed on the repaired area to ensure the entire defect has been removed, and no additional discontinuities have been discovered during weld metal removal. The excavated area shall be welded in accordance with the requirements of this document. Excessive reinforcement or convexity may be removed by grinding to conform to profile requirements (see 5.10.8) without re-welding.

5.12.5.1 Backgouged roots. The joint root of full penetration joints, welded from both sides, shall be backgouged after sufficient welding has been done on one side and before any welding is started on the opposite side. The weld root area shall be backgouged to sound material.

5.12.5.2 Weld repair excavation preparation. Backgouged roots and weld repair areas shall be contoured to produce an excavation which is fully visible to the welder and allows accessibility of the electrode to all surfaces to be welded. The gouged area configuration shall have side walls sloping without sharp breaks. Prior to resuming welding, inspection of the root or excavated repair area shall be performed with the appropriate NDT method.

5.12.6 Crack repair.

5.12.6.1 Crack identification. Certified NDT personnel shall evaluate a crack indication as relevant or non-relevant by determining appropriate NDT method and identify the beginning and ending locations of relevant crack. See 5.12.1 for cracks that extend beyond the weld zone, which is greater than 5/8 inches (16 mm) beyond the weld toe line.

5.12.6.2 Crack arresting holes. Crack arresting shall be used for partial penetrating cracks as determined by certified NDT personnel or welding engineer. An arresting hole shall be drilled within 1/8 to 1/4 inch (3 mm to 6 mm) past each end of the crack and should be approximately 5/32 to 3/8 inches (4 mm to 10 mm) in diameter. Following completion of the repair, the crack arresting hole shall be repaired in accordance with 5.12.6.3.

5.12.6.3 Crack removal. Grinding or other mechanical methods shall be used to remove the crack and be performed in accordance with 5.12.5. Material temperature shall be monitored during crack removal and shall not exceed the maximum interpass temperature in accordance

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with the qualified weld procedure. Cracks starting within 5/8 inches (16 mm) of the weld toe line are weld zone crack repairs and shall be excavated from the end of the crack towards the weld metal in accordance with 5.12.4.2. Cracks starting outside 5/8 inches (16 mm) shall be considered base material repairs (see 5.12.4).

5.12.6.3.1 Crack types. There are three types of cracks: partial penetration, full penetration accessible from both sides, and full penetration accessible from one side.

5.12.6.3.2 Partial penetration crack removal. Crack arresting should be used for partial penetrating cracks as determined by certified NDT personnel or welding engineer. The following is a recommended practice:

a. Excavate the crack to the same angle and root opening tolerances as allowed in the WPS, with a 45° tapered end (see FIGURE 10).

b. Do not exceed 66% of the material thickness during excavation if this thickness is exceeded (see 5.12.6.3.3). Excavation should include any drilled holes.

c. There shall be a 5/16-inch (8 mm) minimum root opening at the base of the excavated area.

d. Examination of the excavation site in accordance with the original approved NDT method shall be used to ensure the crack is completely removed. If no NDT method was originally used for the weldment, the authorized Level III NDT inspector or welding engineer shall make the decision on the method to apply. If the excavation reaches a threshold of 33% of material thickness and shall be repaired as a PJP crack repair joint (see 5.3.1.3).

e. Weld the excavated area in accordance with the requirements of this document. The use of run-off/run-on tabs and removable backing material is encouraged.

f. The repaired crack shall be inspected in accordance with the original approved NDT method. If no NDT method was originally used for the weldment, the authorized Level III NDT inspector or welding engineer shall make the decision on the method to apply.

5.12.6.3.3 Full penetration crack removal accessible from both sides. Use crack arresting process as described in 5.12.6.2 to propagation of the crack. The following is a recommended practice:

a. Starting on side one of the crack repair, follow the process for partial penetration crack removal in 5.12.6.3.2 steps a, c, and e. The depth of excavation shall be > 0.5 T of the plate to be repaired.

b. Once root pass of side one is completed, inspection of the excavation site per the original approved NDT method shall be used to ensure the crack is completely removed. If no NDT method was originally used for the weldment, the authorized welding engineer shall make the decision on the method to apply. The PT Inspection process shall not be used at this time

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without prior approval from TA. Following completion of the side two root pass, this inspection shall be repeated.

c. Passes shall be applied on alternating sides of the crack. Once welding has been completed on both sides, final inspection shall be performed. The repaired crack shall be inspected in accordance with the original approved NDT method. If no NDT method was originally used for the weldment, the authorized welding engineer or authorized Level III NDT inspector shall make the decision on the method to apply.

5.12.6.3.4 Full penetration crack removal accessible from one side. For materials of any thickness that are only accessible from one side, all repairs shall be in accordance with applicable technical bulletin, technical manual, contract, scope of work, or a repair procedure approved by the procuring contract officer. This is a recommended practice.

5.12.7 Loss of shielding during repairs. A loss of shielding in the repair process shall require visual inspection prior to resumption of welding (see 5.9.2.1.3.1). If the portion of the bead generated without proper shielding is unavailable for visual inspection, all weld material shall be removed prior to the resumption of welding. In cases where the point of shielding loss is unknown, all weld material shall be removed prior to the resumption of welding.

5.12.8 Field repairs. Reprocessing of nonconforming material in accordance with approved written repair procedures and operations to address the nonconformity in theater. The purpose of a field repair is to restore functionality to an item being repaired on an as-needed basis. Field repairs are distinguished from other repairs, in that the item, after field repair, is temporarily excused from completely conforming to all the applicable drawings, specifications or contract requirements. The system will be brought back into full conformity once the repaired item reaches a facility that is capable of performing a complete repair. This type of repair shall not be used in manufacturing of new vehicles or repairs at a facility capable of performing a complete repair.

5.12.9 Repair record retention. A repair shall be documented in a repair report and shall be updated with pertinent repair locations, repair process used, and inspection results after the repair has occurred. The entity shall deliver this documentation to the procuring contract officer upon request.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Armor welding covered in this document is intended for military use. Non-government use of this document is authorized.

6.2 Acquisitions requirements. The list of information provided below should be used as a minimum for contract acquisitions and procurement. The information should be evaluated to determine if the requirements should be in the base contract, performance specification or delivered to the government through a contract data requirements list (CDRL) or equivalent.

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a. Title, number and date of this standard.

b. Qualification records (see 4.1).

c. Previous qualification to AWS D1.9/D1.9M Annex A (see 4.4)

d. Additional welding processes (see 4.7.3).

e. Procedure qualification records (see 5.3.1 and A.3.1).

f. Change in gas hose material (see 5.4.4.3).

g. Welding procedure specification (WPS) (see 5.5.1 and A.3.2).

h. Transferring WPS from entity to sub-entity (see 5.5.3.2).

i. WPS revisions (see 5.5.5).

j. Previous welder or welding operator qualifications (see 5.6.6).

k. Requirements for destructive testing of welds (see 5.7.1).

l. Use of an NDT qualification not listed (see 5.8.5).

m. Use of an NDT method not listed (see 5.9.2.4).

n. Prohibited preheat and interpass temperature measuring tools (see 5.10.4.1).

o. Use of proprietary repair procedures (see 5.12.1).

6.3 Standard units of measure. This standard makes use of both U.S. Customary Units and the International System of Units (SI). Throughout the standard, the user will find dimensions in U.S. customary units followed by SI in parentheses ( ), or in appropriate columns in tables and figures. SI units are an estimated value based on U.S. Customary Units.

6.4 Subject term (keyword) listing.

Electrodes
Filler metal
Nondestructive testing
Procedure qualification record
Radiography
Ultrasonic
Weld
Welder
Welding operator

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Welding procedure
Welder qualification
Welding procedure specification

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TABLE I. Weld class crosswalk.

Prior Qualification to AWS D1.9/D1.9M	This Document	New Procedures Qualify To
A – Cyclically loaded and statically loaded structures with elevated inspection criteria	Class II (see 4.6.1 b.)	AWS D1.9/1.9M Cyclically loaded and statically loaded structures with elevated inspection criteria
B – Statically loaded structures	Class III (see 4.6.1 c.)	AWS D1.9/1.9M Statically loaded structures
C – Ballistically loaded structures	Class I (see 4.6.1 a.)	This document

TABLE II. Weld classification decision matrix.

Severity[1] / Susceptibility[2]	Catastrophic	Critical	Marginal
Frequent	**High (Class I)**	**High (Class I)**	**Medium (Class II)**
Probable	**High (Class I)**	**High (Class I)**	**Medium (Class II)**
Occasional	**High (Class I)**	**Medium (Class II)**	**Low (Class III)**
Improbable	**Medium (Class II)**	**Low (Class III)**	**Low (Class III)**

Notes:

1. Severity refers to either the result or consequence of the failure of the weld.
2. Susceptibility refers to the likelihood of damage resulting in the weld failing during operation or high-strain-rate events.

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TABLE III. Titanium grades for armor classes.

Armor Class	Grade/M[1] Number
Class 1	23/M54
Class 2	5/M54
Class 3	5/M53
Class 4	Not applicable
Class 4a	38/M54

NOTES:
1. M numbers are defined in AWS B2.1/B2.1M-BMG "Base Metal Groupings for Welding Procedure and Performance Qualification"

TABLE IV. Recommended titanium alloy filler metals.

	AWS A5.16/A5.16M Filler Material Grade					SAE AMS Filler Material	
Grade/M[1] Number Qualified	ERTi-1	ERTi-2	ERTi-5	ERTi-23	ERTi-38	SAE AMS4954 - UNS R56402	SAE AMS4956 - UNS R56408
1 (M51)	X	X					
2 (M51)		X					
5 (M54)			X	X		X	X
23 (M54)				X	X		X
38				X	X		X

NOTES:
1. M numbers are defined in AWS B2.1/B2.1M-BMG "Base Metal Groupings for Welding Procedure and Performance Qualification"

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TABLE V. Essential variables.

Paragraph		Description of Variables	Process	
			GMAW	GTAW
5.4.2 Base material essential variables	**.1**	Base material specification	X	
	.2	Base material thickness (T)	X	
	.3	Deposition thickness (t)	X	
	.4	Backing or spacer material	X	
5.4.3 Filler material essential variables	**.1**	Electrode and rod classification	X	
	.2	Number of electrodes	X	
	.3	Cold wire and hot wire	X	
	.4	Electrode diameter	±1/64 inches (.4 mm)	±1/16 inches (1.6 mm)
5.4.4 Shielding	**.1**	Flow rate	Any increase of 50% or a decrease of 20%	
	.2	Composition	X	
	.3	Gas hose selection	X	
	.4	Trailing shield	A decrease in the duration of trailing shield of 15%	
	.5	Backing	X	
	.6	Inert gas filled chamber	X	
	.7	Vacuum chamber	Any increase in pressure greater than 20%	
5.4.5 Joint design	**.1**	Joint design limitations	X	
	.2	Joint soundness	X	
	.3	Joint details for PJP	X	
	.4	Backing or spacer strip	X	
5.4.6 Electrical parameters and technique	**.1**	Amperage	±15%	±20%
	.2	Voltage	±15%	
	.3	Travel speed	±15%	
	.4	Wire feed speed	An increase of 20%	
	.5	Heat input	X	
	.5.1	Bead width limitation	X	
	.6	Current type	X	
	.7	Polarity type	X	
	.8	Output type	X	
	.9	Pulsed current	X	
	.10	Transfer mode	X	
	.11	Tungsten electrode	X	
5.4.7 General procedure parameters	**.1**	Welding process	X	
	.1.1	Multiple welding processes used in PQR	A change in excess of what is allowed in 5.3.1.4	
	.2	Welding method of application	X	
	.2.1	Manual and semiautomatic applications	X	
	.2.2	Mechanized, automatic, and robotic	X	
	.2.2.1	Adaptive control	X	
	.3	Position.	X	
	.4	Preheat temperature	Addition or deletion of preheat or increase of temperature by more than 18 °F (10 °C)	
	.5	Interpass temperature	X	
	.6	Mulitpass and single pass welds	X	
	.7	Backgouging	X	
	.7.1	Method of backgouging	X	
	.8	Post Weld Heat Treatments (PWHT)	X	
	.9	Cleaning method	X	

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TABLE VI. Procedure qualification testing requirements for Class I.

Type	CJP	PJP[5]	Fillet[5]
Visual Testing (5.9.2.1)	Yes	Yes	Yes
Recommended NDT[4] (5.9)	RT	PT	PT
Reduced Section Tensile (5.7.2)	2	2	-
Macroetch Testing[1] (5.7.4)	4	3	3
Guided Side Bends[1,2] (5.7.3)	4	-	-
Proofing Projectile Impact Test (Appendix A)	Yes	-	-
Blast Test (Prototype TDP)	No relation to destructive testing or PQR/WPS qualification		
Live Fire (Prototype TDP)			

Notes:

1. For T ≥ 3/8 inches (9.5 mm), side bend specimens may be substituted for root and face bend specimens. When there are markedly different properties between two different base metals and filler metal, guided bends may be substituted with 4 macroetch specimens and shall require volumetric inspection (see 5.7.4). Note, the requirement is only 4 bends or only 4 macros. However, longitudinal bends may be performed. In either case, when T < 3/8 inches (9.5 mm), two root and two face bends are required
2. Item 2 is a recommended practice, except it is required when bend testing cannot be performed in accordance with note 1.
3. A complete Class I PQR/WPS package shall include all mechanical testing, typical PQR documentation and Class I PPIT PQR documentation.
4. See 5.3.1.3.1 and 5.3.1.3.2 accordingly.
5. A CJP shall be required before qualifying a PJP or Fillet

TABLE VII. Procedure qualification positions for plate.

Qualification Testing per Weld Class		Allowable WPS Qualified Positions		
		Groove		Fillet
Weld Class Type	Position Qualified on PQR	Plate	Pipe/Tube	Plate and Pipe/Tube
Plate Groove	1G	F	F	F
	2G	F, H	F, H	F, H
	3G[1]	F, V[1]	F, V[1]	F, V[1]
	4G	F, OH	F, OH	F, OH
Plate Fillet	1F	-	-	F
	2F	-	-	F, H
	3F[1]	-	-	V[1]
	4F	-	-	F,OH

NOTES:

1. Welding prohibitions of 4.10 a. shall apply

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TABLE VIII. Procedure qualification test criteria for fillet welds.

Test weld size in (mm)	Test plate thickness[1]		Sizes qualified		
	T1 in (mm)	T2 in (mm)	Single pass (W_1) in (mm)	Multiple pass (W_2) in (mm)	Plate thickness
1/8 (3)	1/4 (6)	3/16 (5)	≤ 1/8 (3)	-	1/8 in. (3 mm) to Unlimited
3/16 (5)	1/8 (3)	3/16 (5)	≤ 3/16 (5)	-	
1/4 (6)	3/4 (19)	1/4 (6)	≤ 1/4 (6)	≥ 1/4 (6)	
5/16 (8)	1 (25)	5/16 (8)	≤ 5/16 (8)	≥ 5/16 (8)	
3/8 (10)	1 (25)	3/8 (10)	≤ 3/8[2] (10)	≥ 3/8 (10)	
1/2 (13)	1 (25)	1/2 (13)	≤ 1/2[2] (13)	≥ 1/2 (13)	
5/8 (16)	1 (25)	5/8 (16)	≤ 5/8[3] (16)	≥ 5/8 (16)	
3/4 (19)	1 (25)	3/4 (19)	≤ 3/4[3] (19)	≥ 3/4 (19)	
> 3/4 (19)	1 (25)	1 (25)	≤ 3/4[3] (19)	> 3/4 (19)	

NOTES:
1: The material thicknesses for the joint design application observed in production and qualification may be used in lieu of tabulated thicknesses above.
2: The maximum single pass using manual SMAW or the other conventional semiautomatic arc welding processes that can repeatedly achieve fusion and penetration soundness under most circumstances.
3: The maximum single pass using unconventional, mechanized, or automated processes with high deposition and high current density characteristics.

TABLE IX. Procedure qualification thickness limitations.

PQR Test Assembly Thickness in (mm)	Base Metal Thickness Range Qualified in (mm)[2,3]		Weld Deposit Thickness Qualified Maximum in (mm)[1]
1/8 ≤ T ≤ 3/8 (3 ≤ T ≤ 10)	½T[4]	2T	2T
3/8 < T < 1 (10 < T < 25)	½T	2T	2T
T ≥ 1 (25)	½T	Unlimited	Unlimited

NOTES:
1: Excludes face and root reinforcement.
2: 1.1T for GMAW-S
3: If any single pass is greater than 3/8 in. thick, then the base metal thickness qualified is 1.1T
4: For base metal repair (5.3.6.1) minimum thickness shall be 1/8 in.

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TABLE X. Welder and welding operator essential variables.

Essential variable	Requalification of personnel is required when this change occurs	Personnel	
		Welders	Welding operators
Welding process (see 4.7)	Change to an unqualified process	X	X
Position (see TABLE XI and 5.6.4.1)	Change to an unqualified position	X	X
Base material thickness (see TABLE IX)	Change to an unqualified thickness	X	X
Backing	Exclusion of the use of backing	X	X
Stringer vs. Weave Passes (see 5.10.10)	A change from stringer to weave, or vice versa; if not combined in the same test coupon	X	X
Number of electrodes	When going from single electrode to multiple, but not from multiple to single	X	X
Change in method of application	Change from manual, semiautomatic, mechanized, or automatic welding to any other method of application	X	X
Arc voltage control	A deletion of automatic arc voltage control (GTAW only)	-	X
Adaptive control	Addition or deletion of adaptive controls	-	X

TABLE XI. Welder and welding operator positions.

Qualification testing per class		Allowable qualified positions		
		Groove		Fillet
Weld class type	Test position	Plate	Pipe/tube[1]	Plate and pipe/tube[1]
Plate groove	1G	F	F	F
	2G	F, H	F, H	F, H
	3G	F, V[2]	F, V[2]	F, H, V[2]
	4G	F, OH	F, OH	F, OH
	3G[2] & 4G	F, V[2], OH	F, V[2], OH	All[2]
	2G, 3G[2], & 4G	All[2]	All[2]	All[2]
Plate fillet	1F	-	-	F
	2F	-	-	F, H
	3F	-	-	F, V[2]
	4F	-	-	F, OH
	3F[2] & 4F	-	-	All[2]

NOTES:
1: Only qualifies for pipe or tube ≥ 24-in in diameter.
2: Welding prohibitions of 4.10 a. shall apply.

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TABLE XII. Welder and welding operator qualification criteria.

Type of test weld	Nominal[3] T of test plate inches (mm)	Destructive tests[2]	Qualifies welder for Nominal T inclusive inches (mm)	
			Min	Max
Welder groove weld for plate	3/8 (10) ≤ T ≤ 1/2 (13)	2 bends	1/8 (3)	3/4 (19)
	T ≥ 1 (25)	2 bends	1/8 (3)	Unlimited
Welding operator groove weld for plate	1 (25)	2 bends	1/8 (3)	Unlimited
Plug[1]	3/8 (10) or greater	1 macroetch	1/8 (3)	Unlimited
Fillet[1]	1/2 (13)	1 fillet break test and 2 macroetch	1/8 (3)	Unlimited
Pipe[1] or tube[1]	Any	4 macroetch	1/8 (3)	Unlimited

NOTE:

1. A groove weld qualification shall be required prior to qualifying for a plug, fillet, or pipe & tube.
2. Volumetric inspection may be used in place of destructive testing
3. Nominal has a tolerance of +/- .05

TABLE XIII. Minimum NDT in accordance with weld classification.

Class I[2, 3]	
CJP	**PJP**
Volumetric	Surface or Volumetric
Class II[2, 3]	
CJP	**PJP/Fillet**
Surface	Surface
Class III[3, 4]	
CJP	**PJP/Fillet**
Visual	Visual

Notes:

Volumetric = RT, DR, UT, CT
Surface = ET, PT[1]
Visual = VT[2]

1. PT shall not be used for in-process inspection
2. All classes of welds require visual inspection
3. Can be volumetric, as required by the drawing or TDP
4. Can be surface or volumetric, as required by the drawing or TDP

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TABLE XIV. Visual testing acceptance criteria.

Discontinuity Description	Nominal Material Thickness (CJP) (t) / Weld Deposit Size (PJP) (T)	Discontinuity Limit
Arc strikes	All thickness	See 5.12.4.1
Color	All thicknesses	See 5.9.2.1.3.1
Concavity, fillet welds	All thickness	none
Convexity, fillet welds	t ≤ 1/4 in (6 mm)	≤ 1/16 in (2 mm)
	1/4 in (6 mm) < t ≤ 3/4 in (19 mm)	≤ 1/8 in (3 mm)
	t > 3/4 in (19 mm)	≤ 3/16 in (5 mm)
Corner melt	All thickness	≤ 1/32 in (0.8 mm) or 10%t
Cracks	All thickness	none
Crater cracks	All thickness	none
End/Edge melt	All thickness	≤ 1/32 in (0.8 mm) or 10%t
Unspecified/Added welds	All thickness	none
Depth of Fusion	All thickness	meet or exceed requirements of the fabrication document
Incomplete fusion	All thickness	none
Incomplete joint penetration	All thickness	none
Inclusions	All thickness	shall be treated as porosity
Joint mismatch, butt welds	All thickness	≤ 1/32 in (0.8 mm) or 10%t
Meltback	All thickness	none
Missing weld	All thickness	none
Notching	All thickness	none
Overlap	All thickness	none
Porosity, single (max)	All thickness	< 3/32 in (2.5 mm)
Porosity, single (min)		> 1/32 in (0.8 mm)
Porosity, summation		< 3/8 in (9.5 mm)
Porosity, cluster		1 per 4 in (101 mm) of weld
Porosity, aligned		none
Reentrant angle, fillet	All thickness	> 90°
Reentrant angle, groove		> 135°
Skip	All thickness	none
Undercut	t < 1/8 in (3 mm)	≤ 10%
	t ≥ 1/8 in (3 mm)	≤ 1/32 in (0.8 mm)
Underfill, groove welds	All thickness	none

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TABLE XV. Radiographic testing acceptance criteria.

Discontinuity Description	Nominal Material Thickness (CJP) (t) / Weld Deposit Size (PJP) (T)	Individual Discontinuity Limit	Cumulative Discontinuity Limit
Cracks	All thicknesses	None	None
Incomplete joint penetration (CJP)	t < .5 in (13 mm)	3/32 in (2.4 mm)	3/4 in (19 mm)
	0.5 in (13 mm) < t ≤ 1.5 in (36 mm)	5/32 in (4.0 mm)	1 1/4 in (31.8 mm)
	t > 1.5 in (36 mm)	1/3 in (8.4 mm)	2 in (50.8 mm)
Incomplete fusion (CJP)	t < .5 in (13 mm)	3/32 in (2.4 mm)	3/4 in (19 mm)
	.5 in (13 mm) < t ≤ 1.5 in (36 mm)	5/32 in (4.0 mm)	1 1/4 in (31.8 mm)
	t > 1.5 in (36 mm)	1/3 in (8.4 mm)	2 in (50.8 mm)
Incomplete joint (PJP)	1/32 width full length; 1/16 with 4T in 8T weld length		
Incomplete fusion (PJP)	1/32 width full length; 1/16 with 4T in 8T weld length		
Inclusions, nonmetallic per 1-1/2 in of length	All thicknesses		3/8in (9.5mm)
Porosity, summation per 1 in of length	t < .25 in (6 mm)		3/16 in (4.7 mm)
	.25 in (6 mm) < t ≤ .5 in (13 mm)		1/3 in (8.4 mm)
	t > .5 in (13 mm)		3/8 in (9.5 mm)
Porosity, fine pores per inch	t < .25 in (6 mm)	6	
	.25 in (6 mm) < t ≤ .5 in (13 mm)	12	
	t > 1/2 in (13 mm)	14	
Porosity, coarse pores per inch	t < .25 in (6 mm)	3	
	.25 in (6 mm) < t ≤ .5 in (13 mm)	5	
	t > .5 in (13 mm)	6	
Porosity, aligned per 1-1/2 in of length	All thicknesses		3/8in (9.5 mm)
Porosity, cluster (4 or more pores)	t < .25 in (6 mm)		3/16 in (4.7 mm)
	.25 in (6 mm) < t ≤ .5 in (13 mm)		1/3 in (8.4 mm)
	t > .5 in (13mm)		2/3 in (16.8 mm)
Porosity, single (max)	t < .5 in (13 mm)	3/32 in (2.4 mm)	
	.5 in (13 mm) < t ≤ 1.5 in (36 mm)	5/32 in (4.0 mm)	
	t > 1.5 in (36 mm)	1/3 in (8.4 mm)	

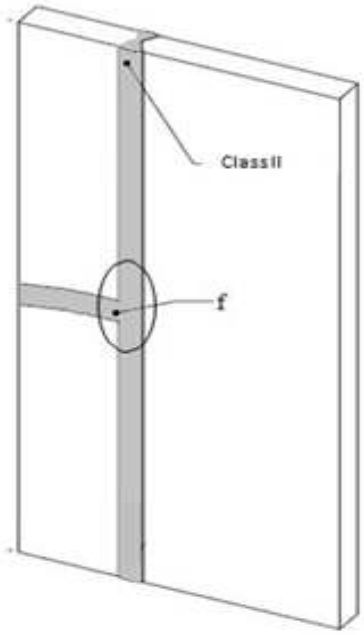
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TABLE XVI. Titanium cleaning critical variables.

Cleaning Method (see 5.11.3)	Description	Cleaning Depth	Fluid Process	Cleaning During	WPS Critical Cleaning Variables																								
					Avg Mill Depth	Air Flow	Pressure	Aggregate / Shot Size	Aggregate Type	Rate in/min	RPM	Wheel dia.	Chemical	Dilution % or PPM	Chemical2	Dilution2 % or PPM	Temperature	Wipe Type	Dwell time	Agitation	Agitation Type	Part Cathode	Part Anode	Voltage	With Water or Rinse	Angle of Incidence	Power	Spot Size
Method 1																													
Method 1.1	Mechanical (Wire Wheel)	Surface		x	x				x	x	x	x																	
Method 1.2	Mechanical (Milling)	Surface		x	x					x																			
Method 1.3	Metal Bead/Shot Blast	Surface		x	x		X	x	x	x																			
Method 1.4	Abrasive Blast	Surface			x		X	x	x	x																			
Method 1.5	Sanding	Surface			x			x	x										x										
Method 1.6	Grinding	Surface			x			x	x		x	x																	
Method 2																													
Method 2.1	Immersion	Contaminant	x										x	x					x	R	R								
Method 2.2	Spray	Contaminant	x										x	x					x										
Method 2.3	Vapor	Contaminant	x										x	x					x										
Method 2.4	Hand Wiping	Contaminant	x										x	x				x	x										
Method 3																													
Method 3.1	Detergent immersion	Contaminant	x										x	x	R	R			x	R	R				x				
Method 3.2	Detergent spray	Contaminant	x										x	x				x	x						x				
Method 3.3	Detergent Ultrasonic	Contaminant	x										x	x	R	R			x	R	R				x				
Method 3.4	Detergent Hot Alkaline or Electrolytic method	Contaminant	x										x	x	R	R	x		x	R	R			R	x				
Method 4	Emulsion cleaning with or without water	Contaminant											x	x											x				
Method 5																													
Method 5.1	Deoxidizing by Immersion	Surface	x										x	x	R	R			x	R	R				x				
Method 5.2	Deoxidizing by Spray	Surface	x										x	x				x	x						x				
Method 5.3	Deoxidizing by Ultrasonic	Surface	x										x	x	R	R			x	R	R				x				
Method 5.4	Deoxidizing by Hot Alkaline Electrolytic method	Surface	x										x	x	R	R	x		x	R	R			x	x				
Method 6	Phosphorous Coat chem conversion	Not used on Ti-Alloy																											
Method 7	Steam cleaning, with or without assisted pressure wash	Contaminant	x				X			x							x		x						R	x			
Method 8	LASER Cleaning	Surface, Contaminant		x	R	R				x																x	x	x	

Note: R is optional - record if used

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Example:

If Class I weld at center section butt joint is done as a sub-assembly and then later gets married to the outer plate details with welds originally designed as Class II, then the intersection about a 2" radius at point f would now be subjected to all the higher classification inspection requirements.

FIGURE 1. Example of weld overlap condition.

ALL WELDING SHALL BE IN ACCORDANCE WITH MIL-STD-3081 CLASS (I, II or II as applicable)

FIGURE 2. Recommended engineering drawing call-out.

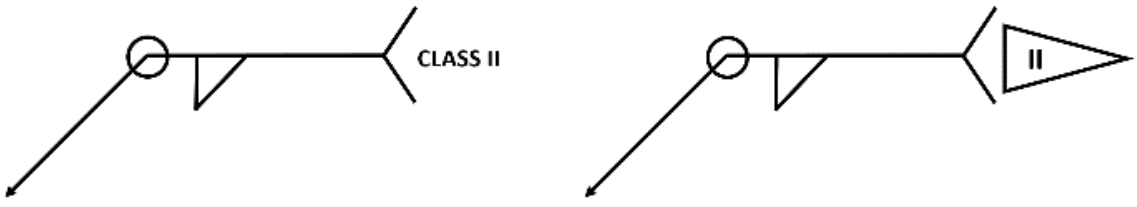


FIGURE 3. Welding symbol classifications.

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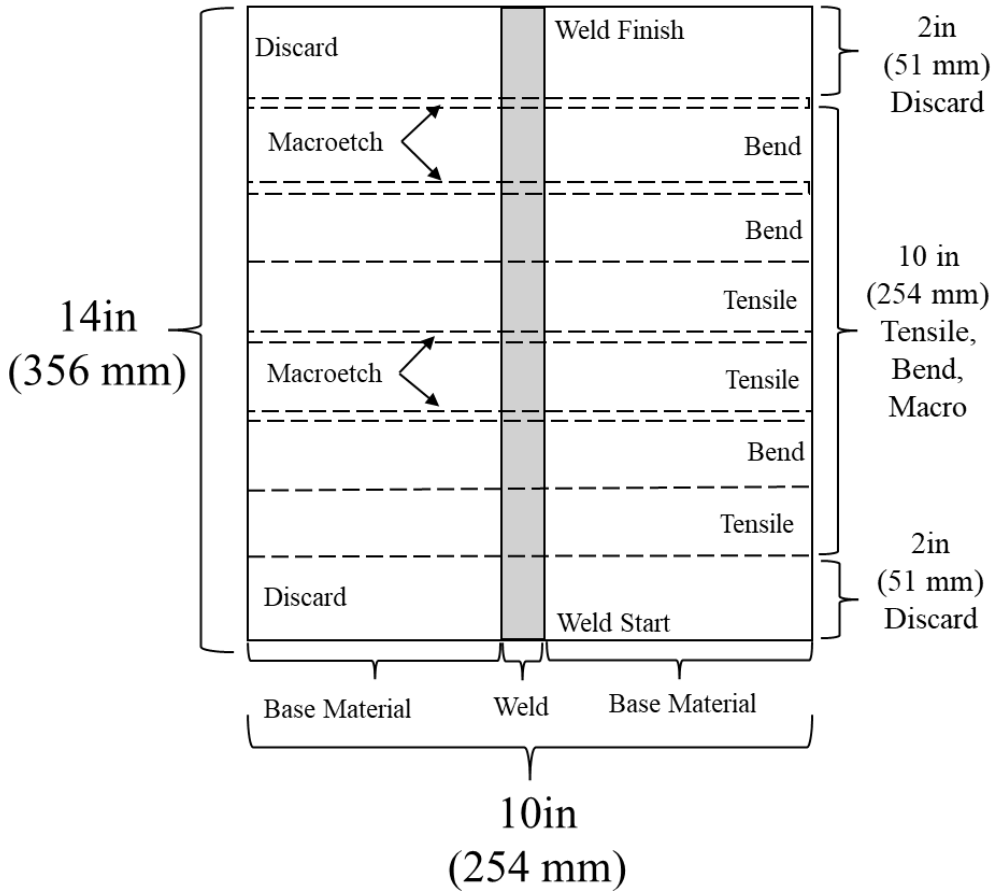
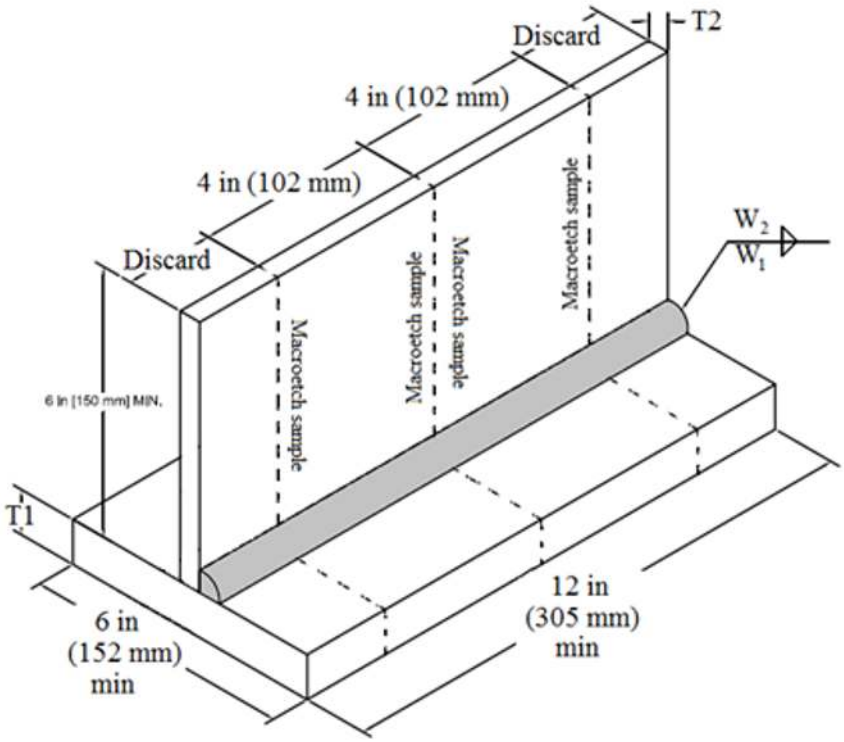


FIGURE 4. Groove procedure qualification test plate.

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Note:

1. Three total macroetch tests shall be completed: each cut end and either side of the center cut.

FIGURE 5. Fillet procedure qualification test plate.

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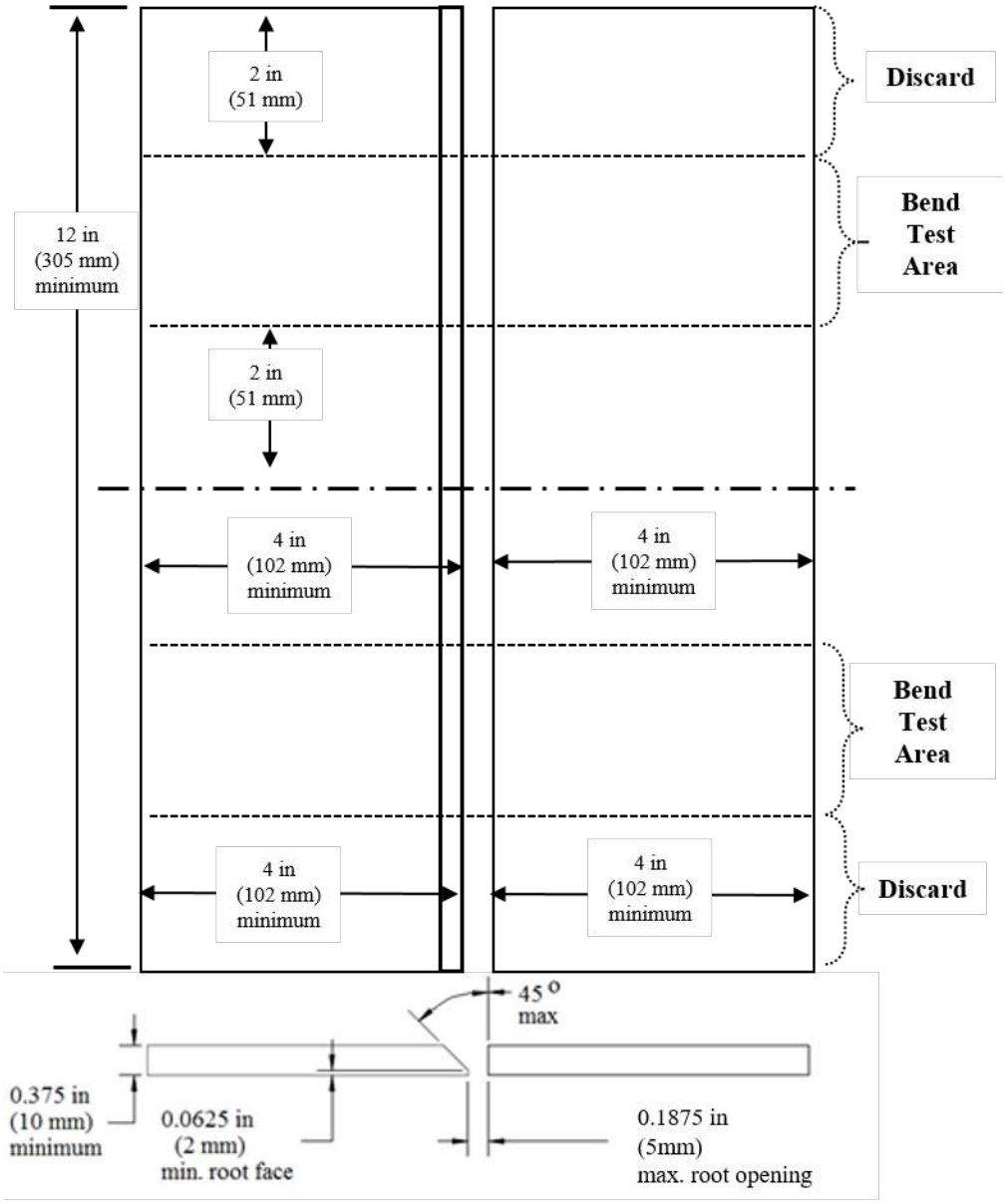


FIGURE 6. Welder and welding operator qualification plate thickness less than 0.5 inches inclusive.

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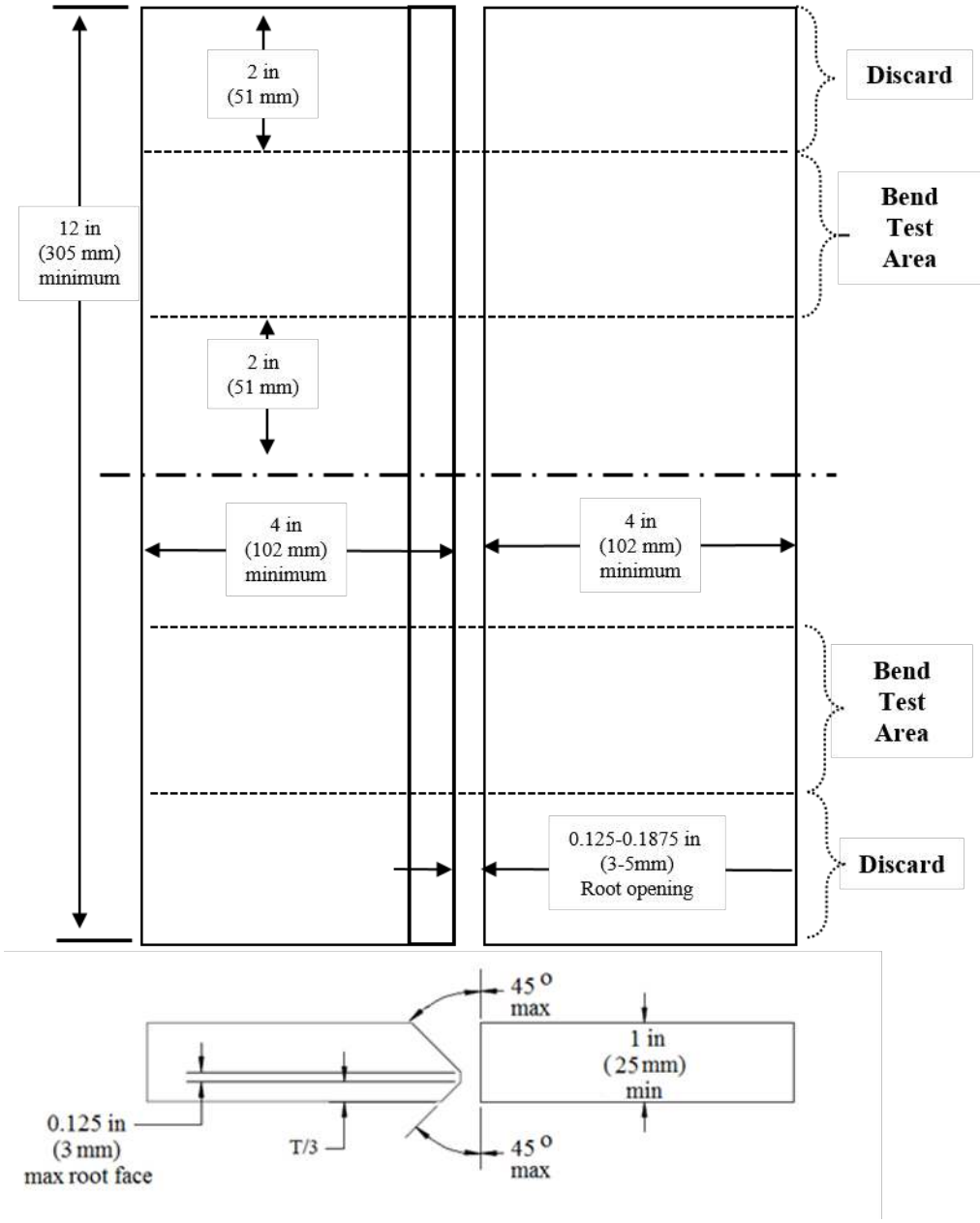


FIGURE 7. Welder and welding operator qualification plate thickness greater than 0.5 inches.

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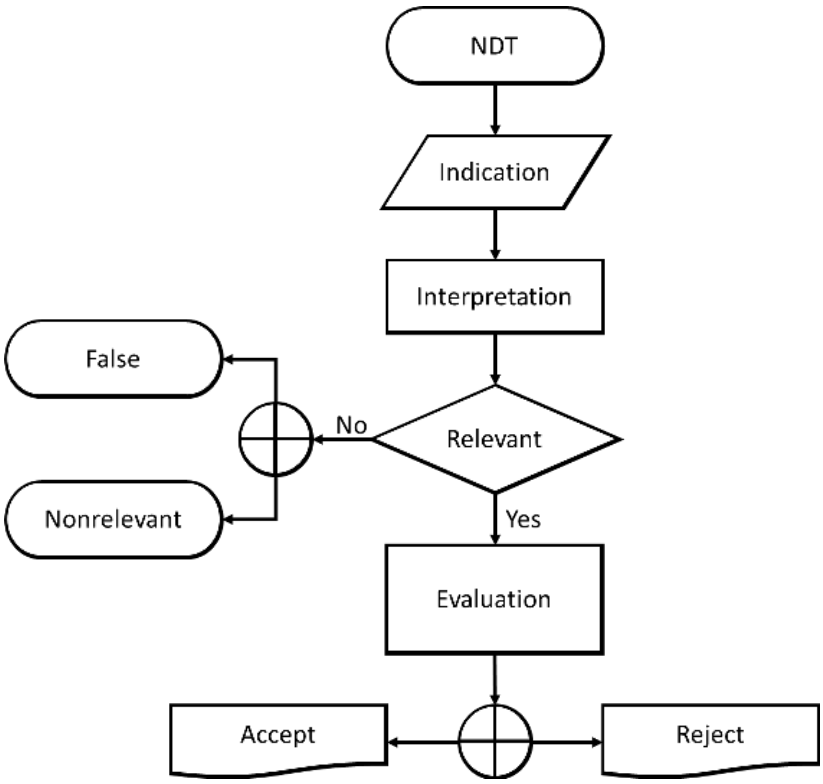


FIGURE 8. Excavated weld metal flow chart.

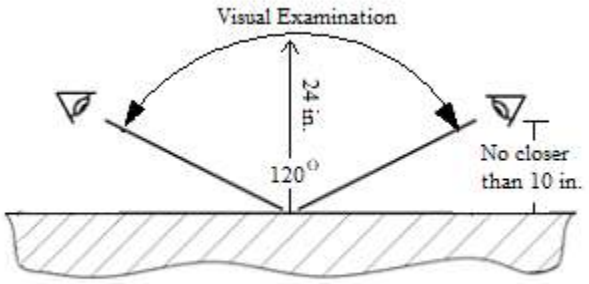


FIGURE 9. Visual inspection.

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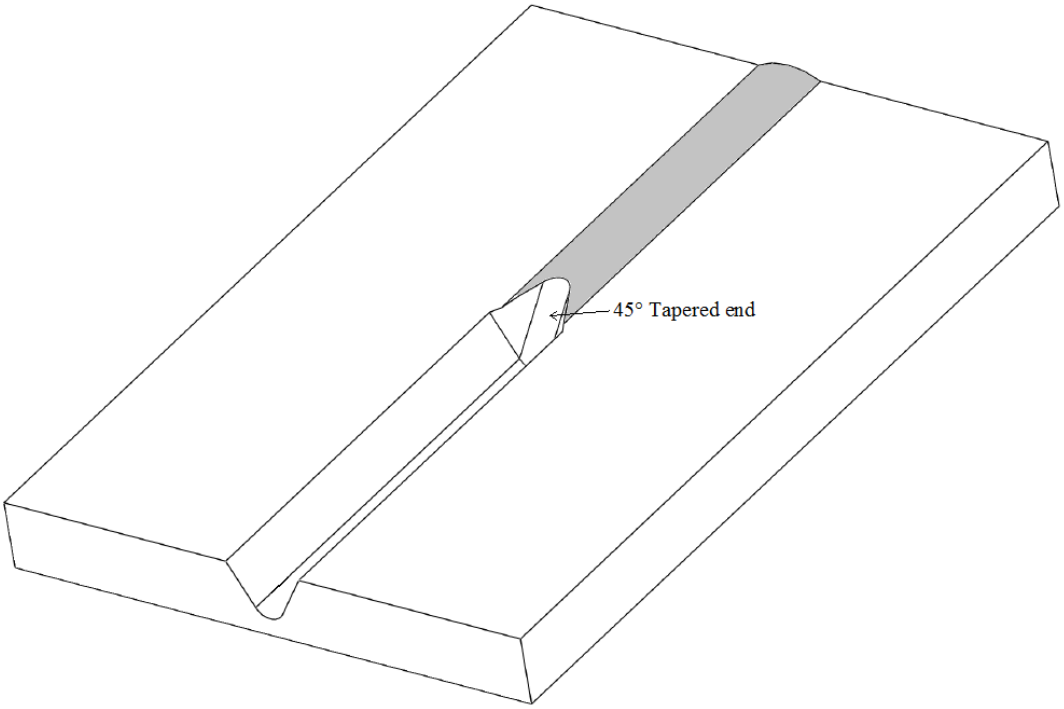


FIGURE 10. Excavated weld metal.

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APPENDIX A: WELDING PROCEDURE QUALIFICATION OF CLASS I WELDS

A.1 SCOPE

A.1.1 Scope. The information contained herein is required for welding procedure qualification of Class I welds.

A.1.2 Application. The PPIT provides validation that a proposed combination of weld process, filler metal, and base material can withstand PPIT. The CJP joints specified in this appendix are intended for use with the PPIT coupons only and do not represent the production representative samples.

A.2 REQUIREMENTS

A.2.1 Requirements. All relevant general requirements (see 4) and detailed requirements (see 5) shall apply to ballistic welding procedures and ballistic welder and welding operator qualification.

A.3 PROCEDURE RECORDS

A.3.1 Procedure Qualification Records (PQRs). Prior to any production, a welding Procedure Qualification Record (PQR) shall be prepared for each ballistic weld in accordance with 5.3, and shall be submitted to the procuring contract officer (see 6.2) for approval prior to usage in production. In addition to destructive (see 5.7) and nondestructive (see 5.9) testing required, each ballistic PQR shall also require PPIT in accordance with this appendix.

A.3.1.1 Previous qualifications. If a previously qualified ballistic PQR is to be used, all requirements of 4.4 and 5.3.4 shall apply. If all requirements are not met, a new PQR shall be developed.

A.3.1.2 PQR base material. The base materials used for ballistic procedure qualification shall meet the requirements of 5.1.2.

A.3.1.3 Welding materials. All welding materials used for a ballistic procedure qualification shall meet the requirements of 5.2.

A.3.1.4 Process qualification essential variables. All required applicable essential variables shall be fulfilled (see 5.4 and TABLE V) for a Class I procedure qualification.

A.3.1.5 Nondestructive testing (NDT). Prior to delivery of the qualification test plate to a Government approved testing agency (see A.5.2), the test plate shall be volumetrically inspected in accordance with 5.9. PPIT shall not be performed until all NDT requirements have been met and all inspection results shall be documented within the PQR.

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A.3.1.6 Destructive testing. Prior to PPIT, all destructive testing requirements of 5.7 shall be met. The test plate may be a separate test plate or may be cut from the I-plate so long as the final dimensions of the I-plate meet the requirements of A.4.1. All testing results shall be documented within the PQR.

A.3.2 Welding Procedure Specification (WPS). Prior to any production, a weld procedure specification (WPS) shall be prepared for each weld in accordance with 5.5 and shall be submitted to the procuring contract officer for approval (see 6.2).

A.4 TEST PLATE

A.4.1 Dimensions of test plate. The size and shape of all plate samples shall be fabricated in accordance with FIGURE A-1.

A.4.2 Plate flatness. All welded I-Plates shall have a maximum out of flatness tolerance of 11/16 in. (17.5 mm) in any direction. Mechanical straightening methods are allowed to meet flatness tolerances.

A.4.3 Test plate thickness. The production thickness ranges from TABLE A-I shall be used to determine the thickness of the test plate required for qualification. Qualification to the minimum thickness and maximum thickness ranges does not qualify any other thickness range unless the ranges of the two qualification procedures overlap. The total number of plates depends on the intended production thickness range.

A.4.3.1 Dissimilar thickness welds. When a production weld is to be fabricated from two dissimilar thicknesses, a minimum of one test plate of each thickness involved shall be prepared. For example, a 1 inch (25 mm) plate welded to a 0.375 inches (10 mm) plate would require two test plates, both a 1 inch (25 mm) and a 0.375 inches (10 mm) test plate.

A.4.4 Test plate armor types. A separate PPIT test plate shall be prepared for each type of armor used for production. Armor is defined as:

a. Armor qualified under different material classes, i.e. MIL-DTL-46077 Class 1 and Class 4 alloys

b. Armor of dissimilar chemistries qualified under MIL-DTL-46077 Class 4

A.4.5 Test plate joint type. Regardless of actual application joint type (CJP or PJP), all test plates shall be fabricated utilizing CJP single or double V groove welds. The test samples shall be Type I-A, Type I-B or Type II (see FIGURE A-2, FIGURE A-3, and FIGURE A-4) in accordance with thickness limitations of A.4.3 and TABLE A-I.

A.4.6 Preparation of test plate. There shall be a unique PPIT test plate for each PQR developed for Class I welds.

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A.4.6.1 Preheat and interpass temperature. Preheat and interpass temperatures shall be in accordance with 5.10.4. Preheat and interpass temperature shall be measured immediately before deposition of each bead in accordance with 5.10.4.1.

A.4.7 Identification marking of test plates. Each PPIT test plate shall be clearly and permanently marked in the upper right corner for easy identification on the front surface of the plate. All markings shall be a minimum of 0.25 inches (6 mm) in height and shall include the plate identification, the manufacturer's name, the contractor's name (if different from manufacturer's name), process and position, and the designation "IMPACT SIDE" to indicate the front of the plate (see FIGURE A-1). All markings shall be fully legible or testing shall not occur. The front of the ballistic plate shall be determined as follows:

a. For Type I joints, the front of the PPIT test plate is considered to be the outside or ballistically exposed surface containing the face of the groove.

b. For Type II joints, the front of the PPIT test plate is considered to be the surface on which welding (excluding tacking) began.

A.4.8 Marking of retest plates. All retest plates shall be marked with the number of the original rejected plate, as well as the new test plate number with the suffix "T" indicating retest.

A.5 PROOFING PROJECTILE IMPACT TESTING

A.5.1 Proofing Projectile Impact Testing (PPIT) requirements. PPIT is required for ballistic welds in addition to the requirements of section 5 of this document. The completed weld procedure for the process used shall be delivered with each PPIT test plate.

A.5.2 Proofing projectile impact testing address. Unless otherwise specified, ballistic testing shall be completed at a certified testing facility. Plates shall be directed to either ATTN: Survivability Armor Ballistic Lab (SABL), 6501 E. 11 Mile Road, RDTA-RS/MS 263, Warren, MI 48397-5000, or Commander, U.S. Army Aberdeen Test Center, 400 Colleran Road. Bldg. 358, ATTN: CSTE-DTC-AT-SL-V Armor Acceptance - B690, Aberdeen Proving Ground, MD 21005-5059.

A.5.3 NDT examination of test plate: Prior to shipping the test plate to the testing facility, all finished welds shall have visual and volumetric examination performed. All testing requirements and acceptance criteria shall be in accordance with this document. Final acceptance testing shall take place in accordance with this document. Should the test plate fail to meet the NDT examination requirements, the PPIT shall not be performed until the defective weld area has been repaired (see 5.12), and able to pass examination.

A.5.3.1 Visual examination of test plates. Prior to PPIT, all completed I-plates shall meet the visual inspection requirements of 5.9.2.1.

A.5.3.2 Volumetric NDT examination. Prior to PPIT, all completed I-plates shall pass a volumetric inspection (see 5.9).

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A.6 TEST RESULTS

A.6.1 Evaluation of test. The test plate shall be tested and meet the requirements in accordance with TOP-2-2-711.

A.6.2 Nondestructive testing after test. All test plates shall be examined with VT (see 5.9.2.1) to indicate a crack and measure the crack length. All test plates may have PT (see 5.9.2.2) performed to assure an accurate evaluation of the crack length if the test director of the testing facility cannot determine the crack length by VT. All PT shall be performed by certified NDT inspector in accordance with 5.8.

A.6.3 PPIT Reports. The testing facility shall document results of the PPIT and provide the results of the evaluation crack length within 10 working days after final NDT after testing is completed.

A.6.3.1 Photographic record. Photographs shall be taken of the entire front and back of the plate as it sits in the test fixture prior to and after any impacts. Additional close-up photographs of the front and back of the impact area showing any cracks present shall also be taken after each impact. Front, back and length of cracking shall be clearly identified for each photographic record. Each photograph shall include an embedded reference scale.

A.6.4 Repair of test plate. Weld repair on a test plate shall not exceed a total length of 8 in. (200 mm) and any defective weld zone shall not be repaired more than once. Repairs shall be to correct discontinuities identified during NDT. No repairs shall be allowed after proofing projectile impact testing. An amendment to the PQR (see A.3.1) shall be submitted with the repaired test plate, and shall include, at a minimum, the following information:

a. The reason for repair.

b. The extent of the repair.

c. The location of the repair.

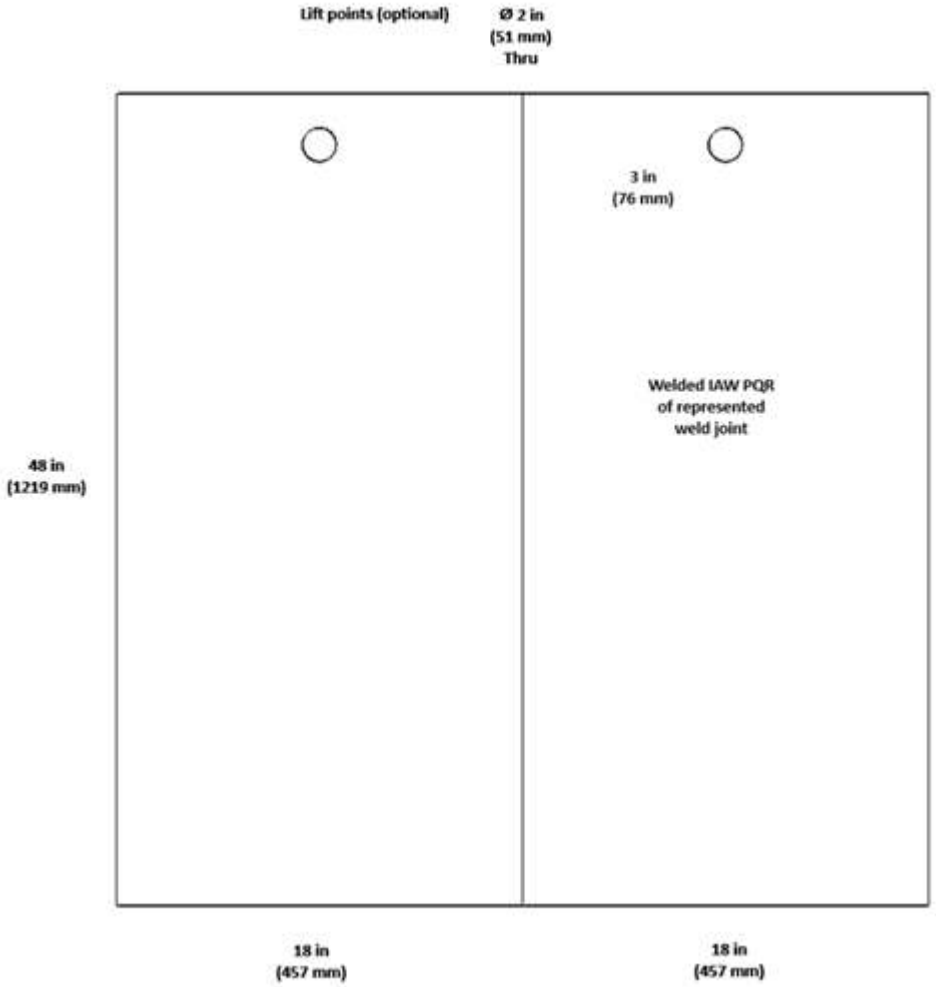
A.6.5 Rejection of test plate. Failure of any test plate to pass either the PPIT, or the second radiographic inspection after allowable repairs, shall be cause for rejection of the recorded welding procedure.

A.6.6 Retest. Retests of a rejected test plate may be made upon the request of the contractor. Two additional test plates shall be fabricated using the same WPS as the previously rejected test plate, marked in accordance with A.4.7 and submitted to the testing facility for retest. Failure of either of these plates shall be cause for rejection of the WPS represented. Only one retest for a welding procedure shall be allowed.

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TABLE A-I. I-test plate thickness and qualification range.

Thickness (T) of I-test plate		Joint type[1]	Thickness (T) range qualified	
In.	mm		In.	mm
0.375	10	I-A or I-B	0.25≤ T ≤0.5	6 ≤T ≤12
1	25	II	> 0.5	> 12



Note:
1. Overall plate dimensions are ±0.5 inch
2. Lift point dimensions are -0 +0.25 inch

FIGURE A-1. Proofing projectile impact testing test plate (I plate).

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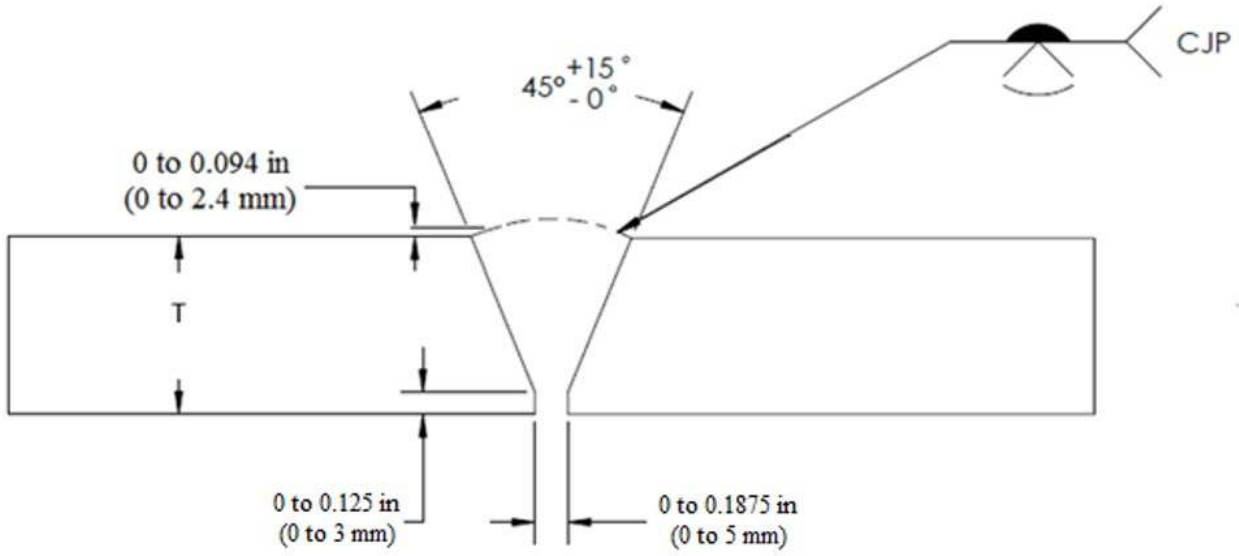


FIGURE A-2. Type I-A, single V-groove butt joint, without backing.

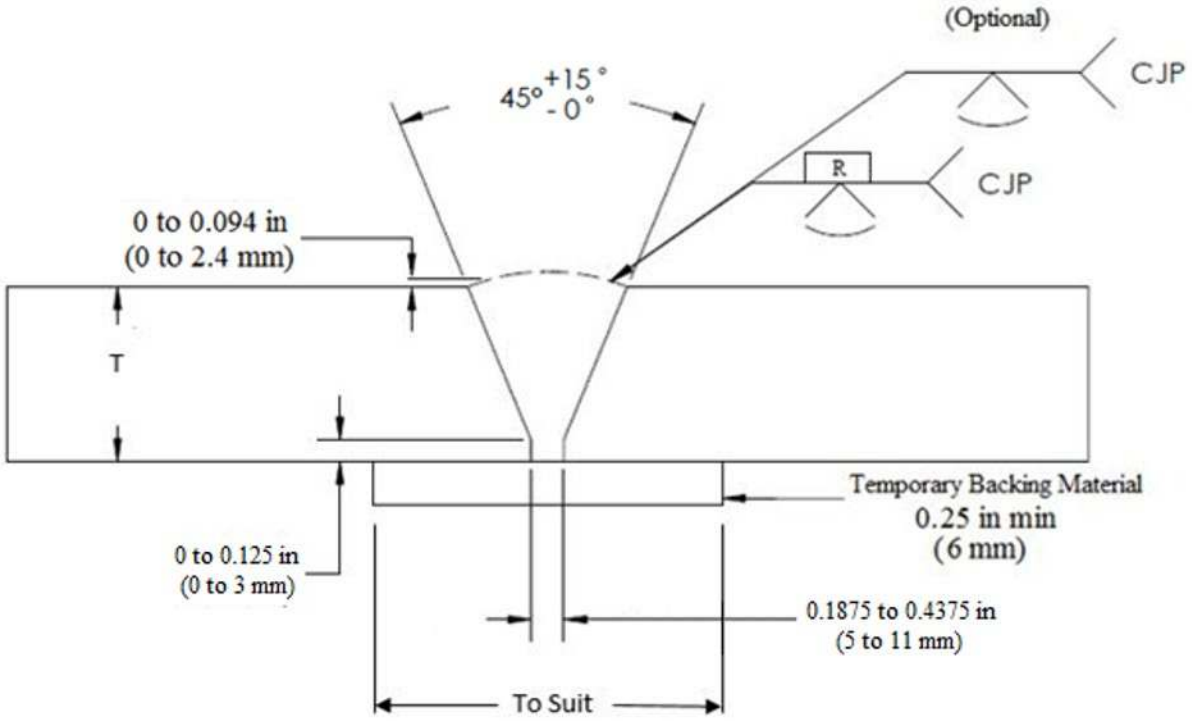


FIGURE A-3. Type I-B, single V-groove butt joint with backing material.

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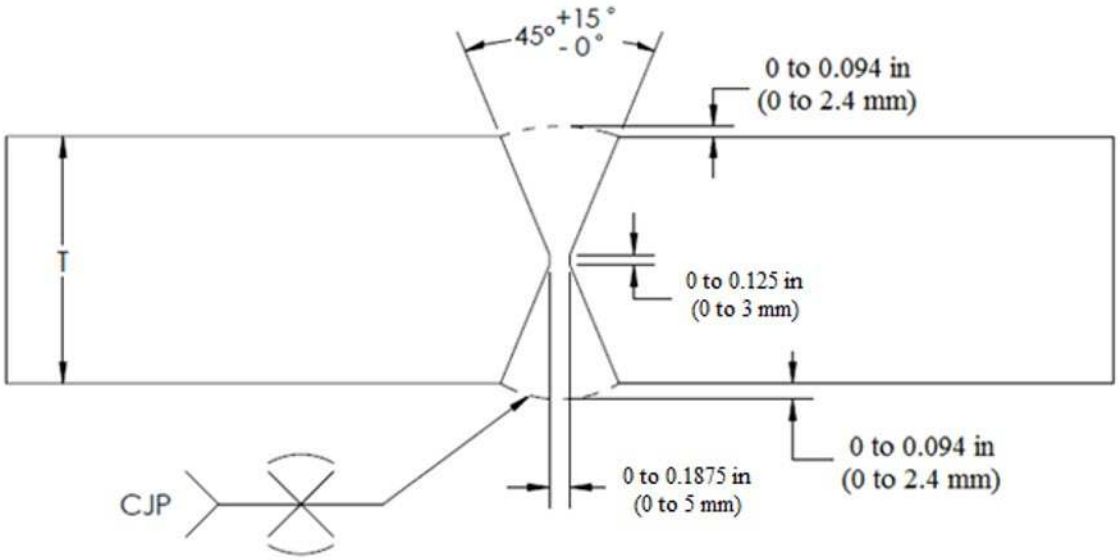


FIGURE A-4. Type II, double V-groove butt joint.

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REFERENCE MATERIALS

B.1 SCOPE

B.1.1 Scope. The documents contained herein are examples of various welding records and forms.

B.1.2 Application. The forms in this appendix may be used in whole or partially as needed.

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PROCEDURE QUALIFICATION RECORD (PQR)-SHEET 1

MODIFIED ☐ QUALIFIED BY TESTING ☐

Identification # _____ Rev. ___ Date _____ By _____
Company Name _____ Performed By _____ Date _____
Welding Process(es) _____ Authorized By _____ Date _____
Equipment Type
Manual ☐ Semi-Automatic ☐
Mechanized ☐ Automatic ☐

JOINT DESIGN USED
Joint Type: Weld Type:
Single ☐ Double Weld ☐
Backing: Yes ☐ No ☐
Backing Material: _____
Root Opening _____
Root Face Dimension _____
Groove Angle _____ Radius(J-U) _____
Backgouging Method (If Yes) Yes ☐ No ☐

POSITION
Position of Groove: Fillet:
Vertical Progression: UP ☐ DOWN ☐
Other _____

ELECTRICAL CHARACTERISTICS
Transfer Mode (GMAW only)
Short- Circuiting ☐ Globular ☐
Spray ☐ Pulsed ☐
Current: AC ☐ DCEP ☐
DCEN ☐
Other _____
Tungsten Electrode (GTAW)
Size _____
Type _____

BASE METALS Base1 Base 2
Material Spec. _____ _____
Type/Grade/Class _____ _____
Thickness: Base1 _____ Base2 _____
Diameter (Pipe) _____

FILLER METALS
Applicable Specification _____
Applicable Classification _____
Applicable F-Number _____

TECHNIQUE
Stringer or Weave Bead _____
Multi-pass or Single Pass (per side) _____
Number of Electrodes
Electrode Spacing Longitudinal _____
Lateral _____
Angle _____

SHIELDING
1. Gas _____ Composition _____
Flow Rate _____
2. Gas _____ Composition _____
Flow Rate _____
Flux Name _____
Electrode-Flux (Class): _____

Contact Tube to Work Distance _____
Peening _____
Interpass Cleaning: _____

PREHEAT
Preheat Temp., Min. _____ Max. _____
Interpass Temp., Min. _____ Max. _____

POSTWELD HEAT TREATMENT
Temp. _____
Time _____

WELDING PROCEDURE

Pass Weld Layer	Process	Filler Metals Class	Filler Metals Dia. (in.)	Current Type & Polarity	Current Amps or Wire Feed Speed	Volts	Travel Speed

Detailed cleaning procedure

Joint Detail

Joint Detail w/ Weld Pass Locations

FIGURE B-1. Procedure Qualification Record (PQR) example.

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PROCEDURE QUALIFICATION RECORD (PQR)-SHEET 2

Test Results

Tensile Test

Specimen	Width (in.)	Thickness (in.)	Area (in.)	Unlimited tensile Load lb.	Ultimate Unit stress, psi	Character of failure and location
#1						
#2						
#3						
#4						

GUIDED BEND TEST

Specimen No.	Type of bend	Result	Remarks

VISUAL

Appearance _____
Undercut _____
Piping porosity _____
Convexity _____
Test date _____

RADIOGRAPHIC-ULTRASONIC EXAMINATION

RT report no.: _____ Result _____
UT report no.: _____ Result _____

Test conductor and interpreter _____ Print
_____ Signature
_____ Title
_____ Date

Stamp Location

ALL-WELD-METAL TENSION TEST

Tensile Strength _____
Yield Strength _____
Elongation in 2 in. % _____
Laboratory tests no. _____

FILLET/GROOVE WELD TEST RESULTS

Minimum size multiple pass = N/A

Macro-etch
1.) _____ 3.) _____
2.) _____

Bend Break
1.) _____
2.) _____

Welders name _____ Clock No. _____ Stamp No. _____

Test Conducted By _____

Test number Per _____

IMPACT TEST

Size of Specimen _____ 1.) ☐ 4.) ☐ Average ☐
Test Temp _____ 2.) ☐ 5.) ☐ High ☐
Ft-lb ☐ Joules ☐ 3.) ☐ 6.) ☐ Low ☐

Signed by _____
Title _____
Date _____

CWI Stamp Location

Signed By _____
Title _____
Date _____

FIGURE B-1. Procedure Qualification Record (PQR) example – Continued.

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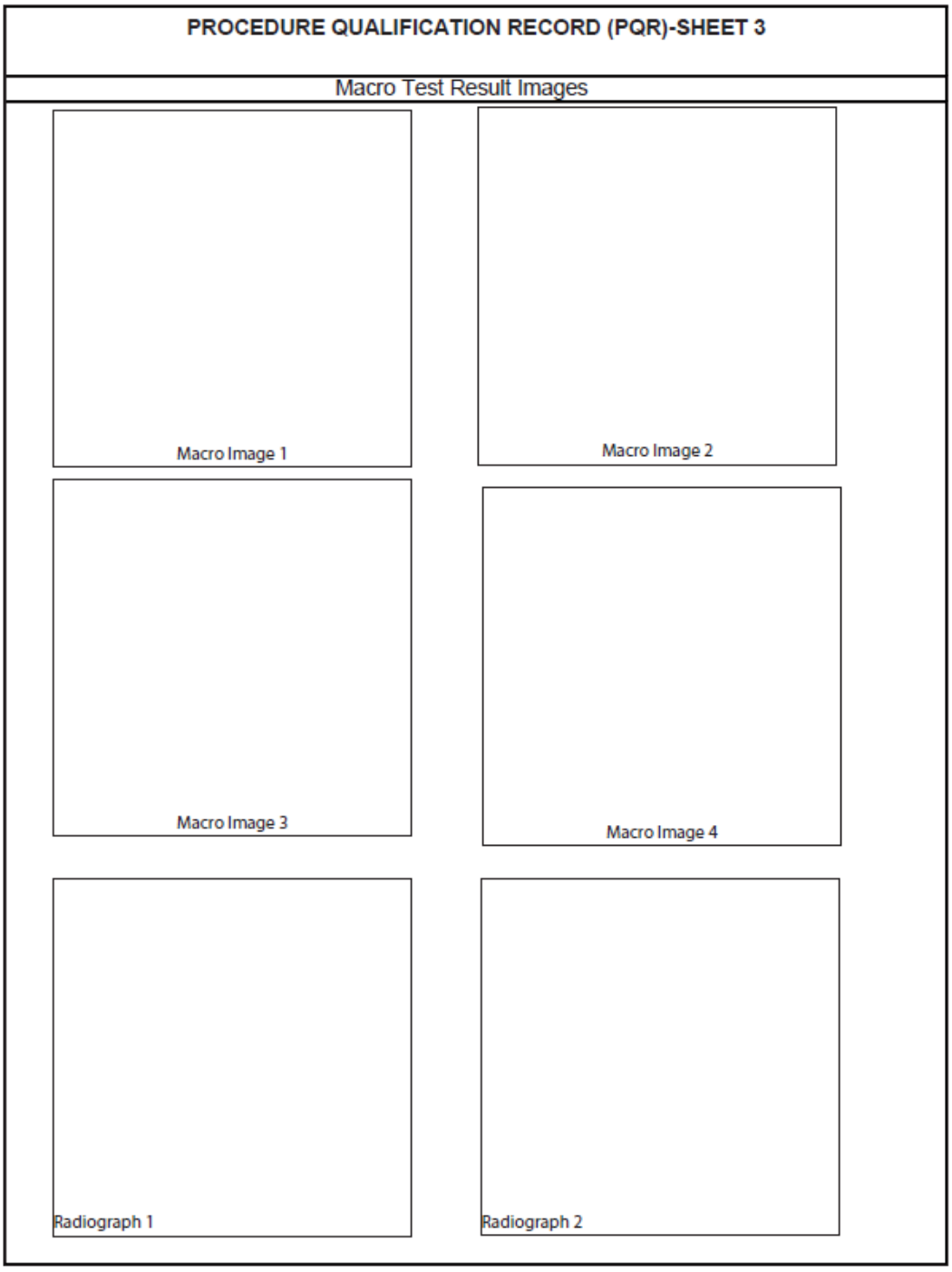
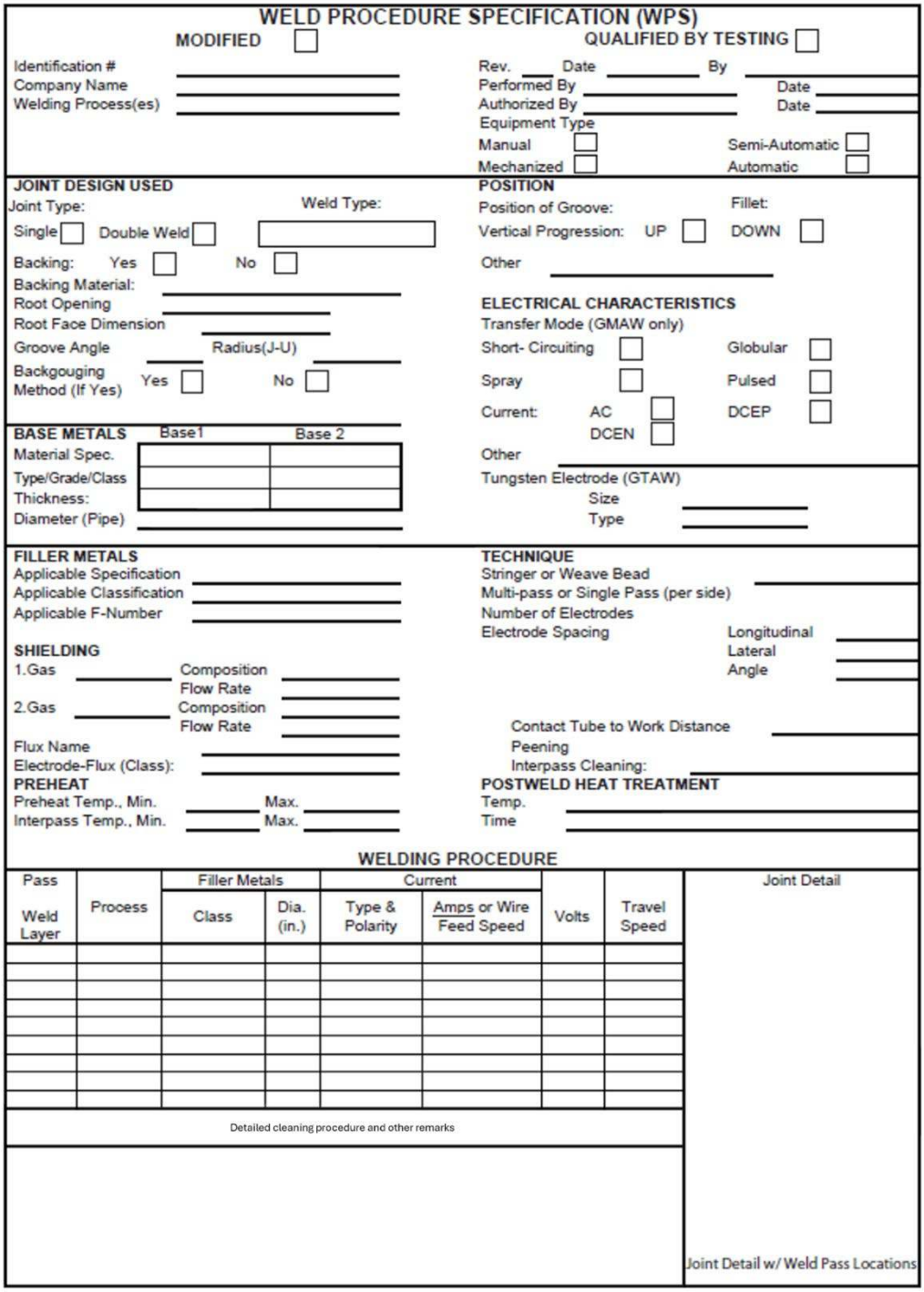


FIGURE B-1. Procedure Qualification Record (PQR) example – Continued.

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WELD PROCEDURE SPECIFICATION (WPS)

MODIFIED ☐ QUALIFIED BY TESTING ☐

Identification # _____ Rev. ___ Date _____ By _____
Company Name _____ Performed By _____ Date _____
Welding Process(es) _____ Authorized By _____ Date _____
Equipment Type
Manual ☐ Semi-Automatic ☐
Mechanized ☐ Automatic ☐

JOINT DESIGN USED
Joint Type: Weld Type:
Single ☐ Double Weld ☐
Backing: Yes ☐ No ☐
Backing Material: _____
Root Opening _____
Root Face Dimension _____
Groove Angle _____ Radius(J-U) _____
Backgouging Method (If Yes) Yes ☐ No ☐

POSITION
Position of Groove: Fillet:
Vertical Progression: UP ☐ DOWN ☐
Other _____

ELECTRICAL CHARACTERISTICS
Transfer Mode (GMAW only)
Short- Circuiting ☐ Globular ☐
Spray ☐ Pulsed ☐
Current: AC ☐ DCEP ☐
DCEN ☐
Other _____
Tungsten Electrode (GTAW)
Size _____
Type _____

BASE METALS

	Base1	Base 2
Material Spec.		
Type/Grade/Class		
Thickness:		

Diameter (Pipe) _____

FILLER METALS
Applicable Specification _____
Applicable Classification _____
Applicable F-Number _____

SHIELDING
1.Gas _____ Composition _____
Flow Rate _____
2.Gas _____ Composition _____
Flow Rate _____
Flux Name _____
Electrode-Flux (Class): _____

PREHEAT
Preheat Temp., Min. _____ Max. _____
Interpass Temp., Min. _____ Max. _____

TECHNIQUE
Stringer or Weave Bead _____
Multi-pass or Single Pass (per side) _____
Number of Electrodes _____
Electrode Spacing Longitudinal _____
Lateral _____
Angle _____
Contact Tube to Work Distance _____
Peening _____
Interpass Cleaning: _____

POSTWELD HEAT TREATMENT
Temp. _____
Time _____

WELDING PROCEDURE

Pass Weld Layer	Process	Filler Metals Class	Filler Metals Dia. (in.)	Current Type & Polarity	Current Amps or Wire Feed Speed	Volts	Travel Speed

Detailed cleaning procedure and other remarks

Joint Detail

Joint Detail w/ Weld Pass Locations

FIGURE B-2. Welding Procedure Specification (WPS) example.

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WELDER/ WELDING OPERATOR QUALIFICATION RECORD (WQR)

Name _____ Identification # _____

Welder Type _____ Date _____

Welding Procedure Specification (WPS) ID# _____

Variables	Actual Recorded Values Used During Qualification	Range Qualified
Process Type		
Automatic/Machine		
Electrode		
Number of Electrodes		
Current/Polarity		
Position		
Weld Progression (Vertical)		
Backing		
Base Material 1 Grade/Class		
Base Material 2 Grade/Class		
Base Material 1 Thickness		
Base Material 2 Thickness		
Base Material 1 Type		
Base Material 2 Type		
Base Material Weld Configuration		
Filler Material Specification # (MIL-E-XXXXX,MIL-DTL-XXXXX)		
Filler Material Class		
Gas/Flux Composition		
Arc Voltage Control		
Joint Tracking Software		

FIGURE B-3. Welder/Welding Operator Qualification Record (WQR) example.

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Welding Procedure Specification (WPS) ID# _____

WQR Test Results

VISUAL INSPECTION

Appearance _____ Pass/Fail _____

Undercut _____

Piping porosity _____ Test date _____

Convexity _____

GUIDED BEND TEST

Specimen No.	Type of bend	Result	Remarks

RADIOGRAPHIC-ULTRASONIC EXAMINATION

UT report no.: _____ Result _____ RT report no.: _____ Result _____

Result _____ Result _____

FILLET WELD TEST RESULTS

Macro-etch Results	Break Test Results	Fillet Size (two locations)
1.) _____	1.) _____	1.) _____
2.) _____		2.) _____

MACRO/RADIOGRAPH IMAGES

Macro Image 1	Macro Image 2	Radiograph 1	Radiograph 2

Test conductor and interpreter

Welders Name _____ Clock No. _____ Stamp No. _____

Test Conducted By _____

Welder Signature _____

Title _____

Date _____

CWI Stamp Location

Test Conductor Signature _____

Title _____

Date _____

FIGURE B-3. Welder/Welding Operator Qualification Record (WQR) example – Continued.

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CONCLUDING MATERIAL

Custodians:
ARMY – AT
NAVY – SH

Preparing activity:
Army – AT

(Project THJM-2026-006)

Review activities:
Army –AV, MI, MR, TE
DLA – DH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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